

Unstructured Data Analysis Using LLMs: A Comprehensive Benchmark [Experiments & Analysis]

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Abstract

The explosion of unstructured data has immense analytical value. By leveraging large language models (LLMs) to extract table-like attributes from unstructured data, researchers are building LLM-powered systems that analyze documents as if querying a database. These unstructured data analysis (UDA) systems differ widely in query interfaces, optimization, and operators, making it unclear which works best in which scenario. However, no benchmark currently offers high-quality, large-scale, diverse datasets and rich query workloads to rigorously evaluate them. We present Bench-U, a comprehensive UDA benchmark that addresses this need. We curate 6 datasets from different domains and manually construct a relational database view for each using 30 graduate students. These relational databases serve as ground truth to evaluate any UDA system, regardless of its interface. We further design diverse queries over the database schema that evaluate various analytical operators with different selectivities and complexities. Using this benchmark, we conduct an in-depth analysis of key UDA components—query interface, optimization, operator design, and data processing—and run exhaustive experiments to evaluate systems and techniques along these dimensions. Our main contributions are: (1) a comprehensive benchmark for rigorous UDA evaluation, and (2) a deeper understanding of the strengths and limitations of current systems, paving the way for future work in UDA.

1 Introduction

Modern organizations store a large quantity of unstructured data such as clinical notes, legal contracts, financial reports, etc., which account for 80%-90% of global data based on IDC research [16]. These vast repositories of unstructured data, if analyzed appropriately, have immense value in various domains.

As an example, healthcare providers often manage a corpus of hundreds of thousands of heterogeneous medical documents, including disease documents (detailing etiology, symptomatology, and progression patterns), drug documents (detailing indications, mechanisms of action, and contraindications) and documents of medical institutions. If there were a powerful unstructured data analysis (UDA) system, it could enable a provider to easily assist a patient who, for example, has symptoms of frothy urine and dull flank pain, by running queries to identify possible diseases and recommend public hospitals within certain distances of the patient’s home that specialize in these diseases.

LLM-powered Unstructured Data Analysis. To support such needs, the database community is actively developing LLM-based systems for UDA [13, 20, 21, 27, 32]. These systems harness LLMs to generate information from multi-modal data lakes (encompassing

text, images, etc.) and perform analytical operations such as filtering, aggregation, and join, thanks to the ever growing semantic comprehension and reasoning capabilities of LLMs.

Although these systems use different query interfaces, e.g., Python or SQL queries, they are all declarative systems, which, similar to relational databases, offer a set of logical operators for users to write a simple program to analyze data. These systems provide optimized implementation of these operators to ensure accuracy and reduce LLM cost. Typically, these systems feature a query optimizer that automatically transforms a user program to an optimized query execution plan, with optimizations such as ordering the filters, converting joins to filters, selecting an appropriate LLM for the task, etc. In this way, these systems abstract away time-consuming engineering details in UDA, while transparently optimizing accuracy and addressing the performance and scaling bottleneck of LLMs.

The Need For a Comprehensive Benchmark. Different systems have distinct implementations of operators and query optimization strategies. Furthermore, evaluations of these systems are often tailored to highlight specific architectural advantages, and therefore fail to provide a comprehensive assessment across diverse scenarios. As a result, it remains unclear which system works best in which scenario. For example, Palimpzest [21] demonstrates its cost optimization capabilities over a dataset of 1000 short emails. However, the short emails can not expose the system challenges regarding text chunking. DocETL [32] evaluates its multi-agent architecture using a single query over the CUAD dataset, which contains 510 lengthy legal contracts. However, the evaluation for DocETL completely ignores the necessity of token cost optimization. SemBench has a distinct and valuable goal: benchmarking semantic query processing over a mix of structured and unstructured data. However, SemBench does not target thoroughly evaluating the capability of the LLM-powered data systems in processing long, complex unstructured documents that raise significant challenges in cost and accuracy.

Therefore, a comprehensive benchmark is still in desperate need to standardize the evaluation of LLM-powered UDA systems and guide future research in this field.

Design Goals. To fill this gap, in this work we propose to construct a benchmark guided by the following design goals:

(1) Realistic Large-scale Unstructured Data. To reflect real-world scenarios and evaluate system scalability and cost efficiency, the benchmark should include large-scale collections of unstructured documents. These documents should be not only numerous but also long-form (e.g., 10K+ tokens), and may contain substantial noise and irrelevant information.

(2) Dataset & Query Workload Diversity for Optimization. Effective optimization in UDA systems must jointly consider the properties

of both datasets and query workloads. Important dataset properties include domain diversity, data modality (e.g., text, images, tables), and document structure. Meanwhile, query workloads should cover diverse analytical operators (e.g., filter, join, aggregation), varied operator compositions and selectivities, and attribute combinations across modalities. These factors together enable systematic evaluation of optimization strategies such as model selection and filter reordering.

(3) High-quality Ground Truth for Fair Evaluation. To enable fair comparison across heterogeneous systems, a benchmark for unstructured data analysis should provide precise and high-quality labels as ground truth. Such ground truth enables consistent and reliable evaluation across systems with different interfaces and execution strategies.

(4) Easy to evaluate existing systems. To comprehend the advantages/disadvantages of existing UDA systems and expose research opportunities, we have to implement them reliably, perform a thorough evaluation, and conduct a detailed analysis of the results.

Our Proposal. We construct Bench-U for unstructured data analysis that meets all the above goals.

Key Insight: A Relational Database w.r.t. Multi-modal Data Lake.

To develop such a benchmark, our key insight is that constructing a relational database from the corresponding multi-modal data lake is sufficient to cover the evaluation needs of all existing systems, including accuracy, latency, and cost.

EXAMPLE 1. Still using the medical application as an example, given a category of files, e.g., disease documents, we can define a relational schema (i.e., a number of meaningful attributes such as disease name, etiology, symptoms, etc.) in advance. Then we extract the values of all attributes from the data lake as the ground truth. In this way, we obtain a relational database consisting of multiple tables, each corresponding to a specific category of files (i.e., disease, drug, medical institution, etc.). Consequently, any analytical queries concerning the aforementioned attributes can be evaluated based on corresponding data records in the constructed database, agnostic to their query semantics and execution strategies. For example, although the semantic operators in Lotus prefer natural language input, it can still compose queries about the attributes covered by the database to evaluate their implementation.

This insight guides us to construct Bench-U that meets all the above goals.

To meet the first design goal, we collect **six** sets of unstructured documents, as shown in Table 2. In terms of the number of documents, all the datasets have at least hundreds of documents and, in particular, Healthcare has 100,000 long and complex documents, which is **100×** more than the existing benchmarks. In terms of the lengths of the documents, three of the datasets have an average length of more than 10,000 tokens. In particular, on Finance dataset, the length of the documents can be up to 838,418 tokens (≈ 100 pages). Effectively processing a large number of long documents that potentially incorporate much noisy information is prohibitively expensive and difficult. This poses significant challenges for document chunking, chunk retrieval, and filtering, which are critical to the performance of UDA systems.

For the second design goal of ensuring the diversity of the datasets and query workloads for effective optimization, we construct datasets from **six** representative domains, including healthcare, law, art, sports, science, and finance. Among them, art and science include images and text, while science additionally includes tables. We construct a workload of **608** queries (**10×** more than prior benchmarks), spanning five categories: Select, Filter, Join, Agg, and Mixed (other complex queries that are combinations of at least 3 operator types). These queries differ in multiple aspects, including different number of filters, various selectivities, binary or multiple joins, w/ or w/o aggregation operations, etc., forming a rich optimization space that enables systematic evaluation of logical and physical optimization techniques such as filter reordering, join planning, and model selection.

For the third goal of fair evaluation, we define **175** meaningful attributes derived from the six datasets, including categorical, numerical, and string types, exceeding existing benchmarks by **5×**. These attributes align the outputs of different systems into a common structured representation, enabling direct and fair comparison across heterogeneous systems. A team of 30 graduate students performs extensive manual annotation, spending in total 4,110 hours (approximately 0.5 hours per document), with the assistance of LLMs in cross-validation for quality assurance.

To achieve the last goal of easy evaluation, we have conducted extensive experiments to evaluate these systems on all datasets, measuring accuracy, cost, and latency across fine-grained query categories to capture differences in system performance. Furthermore, we analyze the experiment results from four perspectives: query interface, query optimization, operator design, and data processing, which are the key building blocks of such systems. This in depth, multi-faceted analysis offer actionable insights for future research.

Contributions. We make the following contributions:

- (1) We present Bench-U, a benchmark for unstructured data analysis that includes large-scale and diverse datasets, as well as a rich set of queries; to the best of our knowledge, it is the first work that constructs a comprehensive benchmark and thoroughly evaluates existing LLM-powered UDA systems.
- (2) We collect six sets of unstructured data from diverse domains, including more than 100,000 documents. We construct 608 queries that involve various analytical operators over the diverse attributes defined in the database.
- (3) We provide comprehensive and cross-validated relational tables as ground truth, annotated by 30 graduate students over 4,110 hours, enabling objective and reproducible evaluation on UDA systems.
- (4) We thoroughly evaluate seven representative UDA systems in accuracy, cost, and latency. Our in-depth analysis offers insights to guide future research in this field.

2 Unstructured Data Analysis Systems

We present the architecture of a typical LLM-powered UDA system, which typically consists of 4 key modules: *the query interface*, *logical optimization layer*, *physical optimization layer*, and *data processing layer*. As shown in Fig. 1, a user submits a query via the interface, which describes the analytical task. The system parses the query into some analytical operators, such as Extract, Filter, Join, and Aggregation. Then, the logical optimization layer determines a logical query plan, such as pushing down predicates

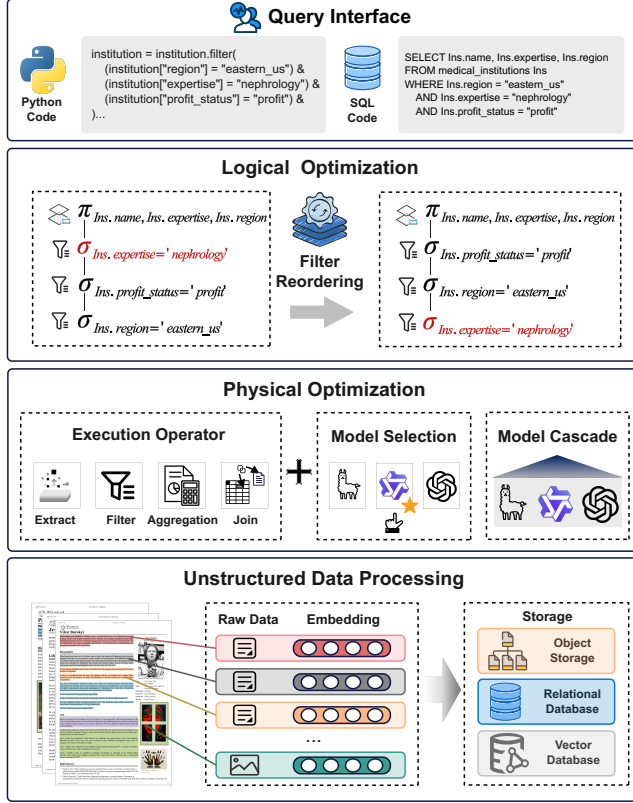


Figure 1: Architecture of Unstructured Data Analysis System. or ordering the operators. Then the physical optimization layer selects the appropriate physical implement w.r.t. each operator in the logical plan, producing an optimized physical execution plan. The data processing layer optimizes the data layout, supporting the aforementioned optimizations. In addition to storing the raw data in distributed storage systems (e.g., Amazon S3), it often segments documents into chunks which are subsequently transformed into embeddings and loaded into a vector database. This enables accurate and efficient retrieval to identify information relevant to a query, speeding up query execution and reducing cost.

Optimization Goals. Unlike traditional databases that focus mainly on optimizing latency, LLM-powered UDA systems have multiple optimization goals, i.e., *accuracy*, *cost*, and *latency*. Accuracy is critical in such systems for two reasons. (1) LLMs are prone to hallucinations, leading to potential inference errors; and (2) to reduce costs, a common practice is to only feed the LLM document chunks highly relevant to the query, rather than entire documents. However, if the retrieval misses relevant chunks, it will degrade the analysis accuracy. On the other hand, the LLM cost and inference latency largely rely on the number of input and output tokens. Reducing them is typically achieved by minimizing the number of input tokens. However, the system still has to ensure that the LLM gets sufficient information to ensure the quality of the analysis.

We detail how existing systems use four modules to achieve the stated goals. Table 1 shows the modules of each system.

2.1 Data Processing

The data processing component handles a wide variety of data types, including plain text, images, etc., which are extracted from

System	Data Processing			Operator				Query Optimization	
	Chunking	Embedding	Multi-modal	Extract	Filter	Join	Agg	Logical	Physical
Evaporate	X	X	X	✓	X	X	X	X	X
Palimpsest	X	X	✓	✓	✓	✓	✓	✓	✓
LOTUS	X	✓	✓	✓	✓	✓	✓	X	✓
DocETL	✓	✓	X	✓	✓	✓	✓	✓	✓
ZenDB	✓	X	X	✓	✓	✓	X	✓	X
QUEST	✓	✓	X	✓	✓	✓	X	✓	X
UQE	X	✓	✓	✓	✓	X	✓	✓	X

Table 1: Overview of Existing UDA Systems.

complex documents such as PDF by OCR tools. Then, it organizes these contents into different formats, which are further processed by different strategies to support downstream analytical tasks.

LOTUS [27], UQE [13] and DocETL [32] typically organize plain text and images into semi-structured files (CSV for LOTUS and UQE, JSON for DocETL), where each entry corresponds to a document. For CSV, plain text is stored in the text column and the paths of the image files are recorded in the image column. For JSON, each document is represented as an object with text and image fields. Subsequently, these systems transform text into embeddings for later semantic analysis, adding the storage path of these embeddings as an additional column in the CSV file. UQE, which supports images, stores the embedding of images as another additional column to support the aggregation operator.

ZenDB [20] and QUEST [33] target analyzing plain text in documents, which is stored in a relational database. Both builds index to speed up retrieval. ZenDB leverages the visual features of PDF such as font size, boldness, and positioning to identify hierarchical section titles and divides the document into semantic units, forming a Semantic Hierarchical Tree (SHT) that reflects the structure of the document. Then, it summarizes the content under each node using the NLTK toolkit [22] and stores the summarization in the node. It calculates the embedding of each sentence to reduce the cost of attribute extraction.

QUEST, on the other hand, constructs two levels of indexes to support accurate and cost-effective attribute extraction. It first generates a summary for each document and encodes it into an embedding to build a vector-based, document-level index. This index allows the system to quickly exclude documents irrelevant to the query. Then, it splits each document into semantically coherent chunks, encodes these chunks into embeddings, and constructs a segment-level vector index, which is used to identify chunks relevant to a to-be-extracted attribute. This avoids feeding the entire document to LLMs to save cost.

Evaporate [5] targets plain text stored in a folder for subsequent analysis. Palimpsest [21] organizes the plain text and images in each document into a directory. Both do not support indexing.

2.2 Query Interface & Operators

Query Interface. Each system provides a query interface for users to define analytical tasks. UQE, ZenDB, and QUEST use SQL-like language to support analysis over unstructured data. For example, in the Healthcare dataset, to find private institutions specializing in nephrology and located in the eastern United States, a user can write a query shown in Figure 1. DocETL, Evaporate, LOTUS and Palimpsest offer declarative Python APIs, corresponding to logical operators in relational databases, for users to compose a query as a Python program, as shown in Fig. 1.

Operators. Bench-U supports 4 common analytical operators.

Extract generates attributes from a set of documents D , formally defined as $\text{Extract}(D, A)$, where A specifies the attributes to be extracted from D and the output is a relational table T_D . For example, to extract the rating and locations of institutions, this operation can be expressed as $\text{Extract}(\text{Healthcare}, [\text{rating}, \text{location}])$. In an SQL-like interface, this corresponds to `SELECT rating, location FROM Healthcare`, while in the Python API, it could be written as `pz.addcolumns(Healthcare, [rating, location])`. In addition, users can provide attribute descriptions as prompts, helping LLMs produce accurate answers. Evaporate employs LLMs to generate code to extract each attribute. Other systems feed the attribute (with optional user descriptions) and relevant document chunks (possibly the entire document) to LLMs for extraction.

Filter. Given a condition C , Filter operation selects a subset of documents that satisfy C from D , denoted by $\text{Filter}(D, C)$. A filter on documents D evaluates whether the relevant attributes of each document satisfy condition C . For example, `lotus.sem_filter('the {document} satisfy {profit_status} is profit')` is a filter in LOTUS. Existing systems adopt two strategies to implement such filters: (1) Palimpzest, QUEST, and ZenDB extract `profit_status` from each document and then evaluate whether it satisfies the filter; and (2) LOTUS, DocETL and UQE take the filter as part of prompts and leverage LLMs to determine whether the filter condition is satisfied. **Join** is a cross-document operation, i.e., combining information from two sets of documents based on a join condition a , formally defined as $\text{Join}(D_1, D_2, a)$. D_1 and D_2 are two subsets of documents, and a is the join attribute. If the system extracts two tables (both contain the attribute a), the tables can be joined on a . For example, from the document subset of disease, the system could extract a table describing the disease and another table with drug attributes from the medication subset; and the two tables can be joined by the disease name. Through a join operation, a user can easily identify possible diseases based on symptoms and then find medications that can treat those diseases. ZenDB and QUEST implement the Join by first extracting the disease table and the medication table, respectively, from two sets of documents and joining the two tables. On the other hand, LOTUS first extracts the disease table and embeds the values of the join key attribute. It then uses the embedding of each disease to retrieve relevant medication documents. From this subset of documents, it extracts the medication table. Finally, the two tables are joined to produce the join result.

Aggregation is defined as $\text{Agg}(D, a, F)$, where a specifies the grouping attribute and F defines the aggregation functions to apply, such as Count, Sum, Avg, Min or Max. This operator supports analytical tasks like “computing the number of institutions grouped by expertise”, i.e., $\text{Agg}(\text{Healthcare}, \text{'expertise'}, \text{Count})$, which can be represented as `pz.GroupBySig(Healthcare, Count, 'expertise')` using the Python API of PALIMPZEST. In implementation, Evaporate, ZenDB, QUEST and Palimpzest extract the ‘expertise’ from all documents, group the values in a table, and then perform aggregation on the grouped data. LOTUS, UQE, and DocETL, on the other hand, first preprocess the documents by clustering them based on their embeddings, and then perform aggregation or batch inference within each cluster as an approximation to save costs. Besides, UQE supports grouping images according to their embeddings.

2.3 Logical Optimization

Given a user-specified query involving multiple operators, UDA systems generate an optimized logical plan to reduce LLMs costs and query latency. The optimizations adopted by existing systems mainly include filter reordering, filter pushdown, and join ordering. **Filter Reordering.** Consider a query that selects artists and their birthdates with two filters, i.e., $F_1 = \text{filter}(D, \text{"lifespan is less than 35"})$ and $F_2 = \text{filter}(D, \text{"tone is warm"})$. Different systems employ different optimization strategies for reordering filters to reduce costs.

(1) *Selectivity-only Strategy.* Palimpzest and UQE adopt a selectivity-only filter reordering strategy that prioritizes a filter with low selectivity. This indicates that the attribute values that a document contains have a small probability to satisfy the predicates of this filter. This reduces the chance of extracting other attributes and evaluating the corresponding filters. These systems estimate the selectivity (denoted by $\text{sel}()$) by sampling. For example, if $\text{sel}(F_1) = 0.2$ and $\text{sel}(F_2) = 0.1$, applying F_2 first would leave $\approx 10\%$ documents for F_1 to evaluate, potentially reducing the cost. However, only considering the selectivities tends to be suboptimal in minimizing the cost. For example, although $\text{sel}(F_2) < \text{sel}(F_1)$, if the document chunks that F_2 has to examine contain much more tokens than those of F_1 , using this order might lead to higher costs than applying F_1 first.

(2) *Selectivity-cost strategy.* To address the above limitation, ZenDB ranks filters based on scores computed by $\text{sel}(F) \times \text{cost}(F)$ and prioritizes those with lower scores. Given the attribute a of a filter F and an unstructured document $d \in D$, it uses $d[a]$ to denote the chunk(s) that are highly relevant to a . The relevance can be computed by measuring the similarity between the embeddings of chunks and the attribute. Let $c(d[a])$ denote the number of tokens of $d[a]$. Then in ZenDB, $\text{cost}(F)$ corresponds to the average number of tokens of chunks relevant to a across all documents, i.e., $\text{cost}(F) = \frac{\sum_{d \in D} c(d[a])}{|D|}$. Therefore, it produces one single filter order w.r.t. all documents, similar to traditional databases. This is suboptimal, as different documents might contain different numbers of tokens w.r.t. an attribute.

QUEST instead produces different orders for different documents considering both their selectivities and costs. Therefore, QUEST does not compute $\text{cost}(F)$ w.r.t. the whole document set D . Instead, for each document $d \in D$, it uses $\text{cost}_d(F) = c(d[a])$ to denote the number of tokens w.r.t. the attribute a in d . Then, it assigns d a specific optimal order which prioritizes the filter with lower values of $\text{sel}(F) \times \text{cost}_d(F)$.

Filter Pushdown. Given a query that joins two sets of documents with filters applied on them, the most straightforward way (ZenDB) is to first pushdown the filters to the two document sets respectively, extract the join key attribute, and then join. Traditional databases adopt this strategy because filters typically have a lower time complexity than joins and thus should be prioritized. However, in this unstructured data analysis scenario where LLM cost is the primary optimization goal, a join may not be more costly than a filter and potentially could have a higher priority. Inspired by this insight, QUEST proposes a join transformation strategy that first extracts the join attribute of one table and then uses the extracted values as filters to filter the other table, i.e., transforming a join into a filter. Treating this automatically generated filter equally to other filters,

QUEST uses the cost model discussed above to order these filters. Therefore, QUEST might prioritize joins over filters to minimize the LLM cost, contradictory to filter pushdown in relational databases. **Join Ordering.** For multi-join, ZenDB and QUEST dynamically and progressively decide the join order during query execution. More specifically, they first select two tables to join based on their cost model, and it will determine the next join only after the first join finishes execution. This process iterates in a left-deep manner until all joins have been executed.

2.4 Physical Optimization

Next, the systems select an appropriate implementation w.r.t. each operator in the logical plan based on the properties of the data and user preferences, i.e., physical optimization. We introduce the typical physical optimizations in UDA below.

Model Selection. For each logical operator, the optimizer selects the most suitable model from a set of candidate based on users’ preference, e.g., accuracy or cost. In Palimpsest, if a user wants to achieve target accuracy while at a relatively low cost, the system prefers a lightweight model for simple extraction tasks to reduce cost, e.g., using GPT-4.1-mini to extract “birthdate” from the Art dataset. Conversely, for complex semantic analysis where Llama fails to meet the target accuracy, it employs more advanced models, such as using GPT-4.1 to infer whether a case is a first-instance trial in the Legal dataset.

Model Cascade. In addition to selecting one model that is the most suitable for the operator, the optimizer in LOTUS applies the model cascade technique to save more cost while meeting the users’ target accuracy. More specifically, it uses a sequence of different models to execute an operator. These models have various characteristics, such as diverse qualities, varying costs, and different levels of latency. For example, the model cascade can be {GPT-4.1-nano, GPT-4.1-mini, GPT-4.1}, which begins with the cheapest GPT-4.1-nano. LOTUS immediately returns the result if the GPT-4.1-nano output meets the target accuracy. If not, the input proceeds to the next model in the cascade, i.e., GPT-4.1-mini, until obtaining a satisfactory output or reaching the last model.

Parallel Execution. Leveraging efficient batched inference with vLLM [18], LOTUS and DocETL process multiple documents concurrently, allowing efficient operator execution over large-scale document collections.

Dataset	#-Domain	#-Attributes	#-Files	Tokens (Max/Min/Avg.)	Multi-modal
Art	1	19	1,000	1,665 / 619 / 789	✓
CSPaper	1	20	200	107,710 / 5,325 / 29,951	✓
Player	4	28	225	51,378/73/8,047	✗
Legal	1	19	566	45,437 / 340 / 5,609	✗
Finance	1	25	100	838,418 / 7,162 / 130,633	✗
Healthcare	3	51	100,000	63,234 / 2,759 / 10,649	✗

Table 2: Statistics of datasets.

3 The Benchmark Construction

In this section, we first overview the construction process of Bench-U and then introduce each step in detail.

3.1 Overview

Bench-U consists of 6 datasets: Art, CSPaper, Player, Legal, Healthcare and Finance. We first collected and pre-processed the raw data. Next, we followed a semi-automatic, iterative process to identify the attributes of the relational tables. We then extract the values

of these attributes from these datasets, i.e., labeling ground truth, combining iterative LLM prompting and human labeling. Finally, we produce a comprehensive set of queries, which can be divided into 5 major categories and 42 sub-categories.

For each dataset, we provide benchmark results in a JSON file, where all documents are decomposed into separate modalities (text, tables, and images). Each document maps to a field in the JSON file, and the different modalities are stored in distinct entries. In this way, if a user wants to test her system using Bench-U, she can directly download the processed data, load the data into the system, run her queries, and compare the results with the ground truth table stored in the relational databases.

3.2 Datasets

Bench-U consists of six datasets with their statistics summarized in Table 2. Next, we describe these datasets below. For more details, refer to our website.

Art is collected from WikiArt.org [7], which covers artists and their artworks spanning from the 19th to the 21st centuries. Each document corresponds to an artist including biographical information, artistic movement, a list of representative works, and an image of a representative work.

CSPaper dataset is crawled from Arxiv[1] containing 200 research papers annotated with key attributes, including authors, baselines and their performance, the modalities of experimental datasets etc. In particular, some papers describe the performance of all baselines in the main text, while other papers only describe the best-performing baselines and leave other results in tables or figures, resulting in an analysis scenario with mixed-modal.

Player dataset is crawled from Wikipedia[24] that contains information about basketball including players, teams, team managers, etc., from the 20th century to the present, covering their basic and statistic information.

Legal is sourced from AustLII [8] with 570 professional legal cases from Australia between 2006 and 2009, covering different types such as criminal and administrative. Each case document typically includes evidence, charges, legal fee, etc.

Finance are collected from the Enterprise RAG Challenge [14], containing annual and quarterly financial reports published in 2022 by 100 listed companies worldwide with an average token length of 130,633. Each record typically includes mixed types of content like company name, net profit, total assets, etc.

Healthcare is obtained from MMedC [28], with healthcare documents since 2020. This dataset contains a massive amount of files (100,000), each file having 10,649 tokens on average. It covers various types of healthcare information, like drugs, diseases, medical institutions, news, interviews crawled from large-scale web corpora and open-access healthcare websites.

Summarization. These datasets show various characteristics. Compared to other datasets, the Art dataset is less complex due to its short documents and well-defined structure, but each document contains a corresponding image. CSPaper is a moderately complex academic dataset composed of research papers that contain text, tables, and images. Legal is more complex because it is a domain-specific dataset containing multiple attributes that require semantic reasoning to extract. Finance is another complex domain-specific dataset. The key challenge it introduces is the length of the

documents – up to 100 pages. Lastly, Healthcare has the largest number of documents, containing rich information. Healthcare and Player contain multiple categories of files, e.g., disease, drug, medical institution, suitable for evaluating join queries.

Note that previous works [5, 10] also use the sources of Player, Art, and Legal in their experiments. We largely augment them by adding new documents, attributes and comprehensive query workloads, addressing the issue that the original datasets were curated primarily for extraction and offered limited query coverage. **Data processing.** We design a unified data preprocessing pipeline to handle datasets with various features. For each dataset, we first collected the raw data and utilized the MinerU [37] toolkit to parse the data when dealing with complex formats such as PDF (e.g., the Finance dataset). Then, we organized the dataset into a JSON file, where each object corresponds to an unstructured document with multiple fields including text, image URL, and metadata. In particular, for the Healthcare and Player datasets, we divided the documents into multiple categories with different topics, each of which corresponds to a relational table. For Healthcare, it originally has 680,000 documents on healthcare information, from which we sampled 100,000 ones. Then we used LLMs to read these samples and identified three major categories of files (6,100 documents in total), i.e., disease, drug, and medical institution. Note that we only define and extract attributes over the three major categories. Other documents are retained to create a realistic large-corpus setting, where only a subset of documents (the above three categories) is relevant to a given query. This allows our benchmark to evaluate the retrieval capability of UDA systems. For Player, we identify 4 subsets of documents belonging to different categories, i.e., players, teams, team managers, cities, amenable to evaluate join queries.

3.3 Ground Truth Labeling

To label the ground truth, we first identify a number of significant attributes from each dataset and then manually extract their values. **Attributes Identification.** We hire 6 Ph.D. students from different majors (e.g., finance, law, medical) in our university to read these documents carefully and identify attributes that pose different levels of challenges to extract. For example, the attribute Judge_name is easily identifiable, since it is usually located at the beginning of each document in Legal. In Finance, Business_cost is the sum of several cost components, which vary across industries—for example, raw materials and wages for car manufacturers versus product sourcing and logistics for supermarkets. A labeler must identify the relevant components of Business_cost from the document context, extract their values, and aggregate them into the final Business_cost. In addition, image files also contain easy to extract attributes like Tone, whose answers often fall into “Neutral”, “Bright” and “Dark” that can be easily categorized. Difficult attributes, such as “Style”, require art expertise to reliably extract. Furthermore, to rigorously evaluate cross modal reasoning, we design complex attributes that require joint analysis across text, tables, and images. For instance, in CSPaper, extracting the most critical ablation component requires systems to first comprehend the textual ablation setup and then visually navigate to specific tables or charts to compute which component removal causes the most severe numeric drop.

Labeling. To ensure high-quality ground truth for Bench-U, we hire a total of 30 graduate students to manually label these attributes, spending approximately 4,110 human hours. To be specific, for each document, an ensemble of diverse LLMs independently retrieves candidate text chunks for each attribute. Human labelers then review the union of these retrieved chunks to manually extract the corresponding values as ground truth. Note that the labeler uses LLMs as an assistant. If the labeler cannot extract the attributes from the retrieved content, she will manually read the document and provide the answer. Importantly, during extraction, we also ask humans to record the chunks where each attribute is extracted so that users can verify the source of every labeled value. Moreover, to further ensure labeling quality, we use LLM-as-a-judge to verify the accuracy of the human-labeled results. Whenever the LLM flags an extracted attribute as incorrect, we ask humans to double check the corresponding ground truth.

However, for the large-scale dataset Healthcare, it is impractical to manually label all the ground truth. Therefore, we adopted a semi-automated iterative labeling strategy. Recap that Healthcare contains a large number of documents that belong to categories other than the three major ones mentioned above. Therefore, these documents rarely include entities in the major categories. Consequently, we asked LLMs to analyze 100,000 documents to label whether each of them belongs to the three categories. If so, we ask humans to label attributes; otherwise, the ground truth is NULL. Remark that the LLM ensemble serves exclusively as auxiliary tools for retrieval, while humans label the ground truth. This ensures the ground truth reflects objective facts rather than LLMs outputs, preventing the benchmark from inheriting biases specific to certain prompting strategies or models.

Quality Assurance. We conducted an explicit Inter Annotator Agreement (IAA) evaluation to ensure labeling quality. We randomly sampled 191 documents across all domains and required three independent annotators to extract attributes, resulting in 3,445 extracted attribute values for agreement analysis. We employ the pairwise agreement ratio [6] ($S = \frac{N_{agree}}{N}$). Accordingly, N_{agree} is defined as the number of instances where two annotators extract the same value for a specific attribute within a document, while N represents the total number of corresponding annotator pairs across all 3,445 cells. The evaluation demonstrates strong agreement, with scores ranging from 0.88 on highly specialized domains (Finance and Healthcare) to 0.98 on general domains (Player). Furthermore, we find that the annotation inconsistencies primarily stem from varying levels of domain expertise rather than factual errors. In specialized fields, annotators who lack relevant expertise may systematically misinterpret complex implicit logic (e.g., deducing whether a legal case is a first instance trial).

3.4 Query Construction

As shown in Figure 2, we divide the queries into five main categories based on the primary query operators, as different operators introduce different technical challenges: (1) Select, involving only Extract operations; (2) Filter, combining Extract with Filter operations; (3) Join, involving Extract and Join operations; (4) Agg, incorporating Extract with Aggregation operations; and (5) Mixed, including several categories of operators listed above. Note that every query includes an Extract step to return the attributes

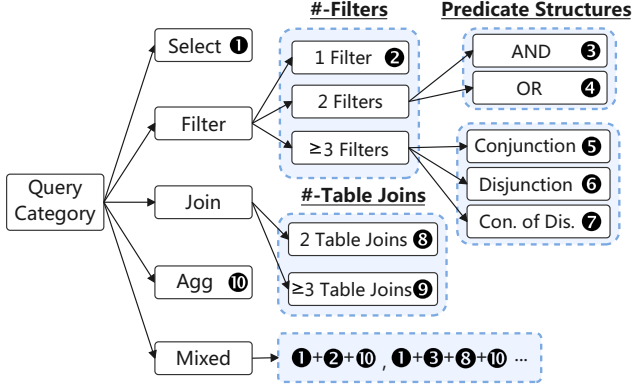


Figure 2: Query Categories.

to be analyzed. We then further decompose each main category into subcategories to construct a broad range of queries that differ in modality, filter complexity in terms of the number of filters, predicate structures (the combination of conjunction or disjunction), selectivities, single or multiple joins, w/ or w/o aggregation, etc. These queries are in the form of both SQL-like queries and Python code, thus able to test systems with different interfaces, like ZenDB and Palimpsest. An example is shown below.

```

1 SELECT {Attribute}(s),
   {agg_func}({Attribute}(s))
2 FROM diseases
3 JOIN drug ON disease.name = drug.disease
4 WHERE {Attribute}{operator}{literal}
5 GROUP BY {group-by};

```

A SQL subcategory Example.

```

1 disease_doc = disease
2 drug_doc = drug
3 disease_doc = Filter(disease_doc,
   {Attribute}{operator}{literal})
4 disease_drug = Join(disease_doc, drug_doc,
   disease_doc.name = drug_doc.disease)
5 result = Extract(disease_drug,
   {Attribute}(s))
6 result = Aggregate(result, {group-by},
   {agg_func}({Attribute}(s))

```

A Code subcategory Example.

(1) *For the Select category*, for example, a query such as “SELECT {Attribute}(s)” represents a Select operation used to extract attributes. Here, “{Text Attribute}(s)” serves as a placeholder that can be instantiated with any number of available attributes with different modalities, which are randomly selected to generate concrete queries. These Select queries can clearly evaluate the system’s capability of extracting attributes with different modalities.

(2) *For the Filter category*, we further classify it by the number of filters (one filter, two filters, and more than two filters) and the way of combining the filters (conjunction only, disjunction only, and conjunction of disjunctions), resulting in a total of 6 Filter sub-categories. With varying complexities, these sub-categories of

queries offer a large space for cost optimization (e.g., more filters bring a large optimization opportunity for filter reordering). As shown in the example, the pattern “{Attribute}{Operator}{Literal}” serves as the placeholder format for filter predicates. During query generation, we randomly sample both the filter attribute and the comparison operator (such as $\leq$, $=$, or $\geq$) with equal probability, and we further sample literal values to produce varying selectivities.

(3) *For the Join category*, to thoroughly evaluate the capacities of different systems in optimizing join queries, we curate two sub-categories of join queries, i.e., binary joins and multi-table joins, where more tables incur higher computation cost. During query generation, we explicitly define a join graph w.r.t. each dataset. For example, in Healthcare, valid join paths include Disease $\bowtie$ Drug, along with their corresponding join keys (e.g., Disease.name = Drug.disease). Similarly, in Player, we have Players $\bowtie$ Teams $\bowtie$ Managers, together with join keys that link these tables.

(4) *For the Agg category*, during query generation, we randomly sample the group-by attribute, the aggregated attributes, and the aggregation functions (such as Count) to populate the corresponding placeholders (i.e., {agg_func}({Attribute}(s))).

(5) *For the Mixed category*, it combines multiple types of operators in one query, e.g., first joining two tables and then run aggregation on the join results. It includes 32 subcategories, each of which is combined with at least three of the aforementioned main categories. The above example illustrates one such combination, i.e., selecting attributes with one filter, an aggregation and a 2-table join.

We then apply these subcategories to each dataset to construct queries. After generating a large pool of candidate queries, we ask human experts to select those that are both meaningful and representative. Specifically, for the first four main categories, we retain 10 queries per subcategory in each dataset. Since the Mixed category aims to evaluate the performance of combining different operators, a single query per template per dataset is sufficient. Details of the queries w.r.t. each dataset can be found in the website.¹ Moreover, we have open-sourced the query generation code so that users can create queries on their own.

Workload Characterization. To further provide a fine-grained understanding of the benchmark workload, we characterize its statistical properties and execution behaviors from three aspects.

(1) *Schema Summary:* The constructed benchmark organizes unstructured corpora into relational schemas, where each table contains 5 to 26 attributes, with a total of 175 attributes across all datasets. Datasets such as CSPaper, Finance, Legal, and Art are modeled as single tables, while Player and Healthcare adopt multi-table schemas (4 and 3 tables, respectively), introducing realistic relational structures and join relationships.

(2) *Workload Properties:* Our query workload exhibits diverse selectivity, join cardinality, and aggregation complexity. Specifically, each query contains 1, 2, or ≥ 3 filters (following a 3:4:3 ratio), and varying combinations of predicates result in both highly selective and low-selectivity queries. The selectivity of filter predicates ranges from approximately 1% (i.e., only a small fraction of documents) to over 50%, covering a broad spectrum of selectivity scenarios. For joins, the workload includes multi-way joins spanning 2 to 4 tables, with relationships ranging from near 1:1 to highly skewed

¹<https://bit-datalab.github.io/bench-u-page/>

1:N patterns, leading to diverse join cardinalities and potential intermediate result expansion. For aggregations, the number of groups varies depending on the queried attributes, typically ranging from a few groups (e.g., around 5) to more than 20 groups.

(3) Input–Output Behavior: We further characterize the input and output sizes across query categories based on their operational semantics. For *Select* queries, the output size depends on the number of documents and extracted attributes. For *Filter* queries, the output size decreases according to predicate selectivity. For *Join* queries, the output size depends on the number of joinable records and may significantly increase due to one-to-many relationships and multi-way joins. For *Agg* queries, the output size is typically compact and determined by group cardinality. Importantly, *Mixed* queries exhibit substantial input–output skew: they often start with large inputs, generate expanded intermediate results due to joins, and finally produce compact outputs after filtering and aggregation, reflecting realistic workload patterns in unstructured data analysis systems.

4 Evaluation

We evaluate existing systems on Bench-U to answer the questions:

- **RQ 1:** What is the overall performance of different systems on the benchmark? Which system achieves the best performance under different scenarios?
- **RQ 2:** What is the accuracy of different systems on the benchmark?
- **RQ 3:** What is the cost of different systems on the benchmark?
- **RQ 4:** How efficient are different systems on the benchmark?
- **RQ 5:** How do different logical optimization strategies perform?
- **RQ 6:** How do different physical optimization strategies perform?

4.1 Experimental Settings

Systems for Evaluation. Our benchmark evaluates 7 existing UDA systems. (1) *Evaporate* is a table extraction system, so we employ *Evaporate* to extract structured tables from documents and subsequently execute SQL queries on the resulting tables. (2) *Palimpzest* provides Python API-based operators for unstructured data processing. We convert each SQL query into the corresponding *Palimpzest* code, execute it and obtain the results. (3) *LOTUS* also provides an open-source Python library, which we use to execute queries through its interface. (4) *DocETL* is an open-source project using Python code. We rewrite our queries with *DocETL* library and execute the Python programs. (5) *QUEST* provides SQL-like query interface for processing unstructured documents, and we directly use their code to execute queries. (6) *ZenDB* does not provide their code; therefore, we implement their SHT chunking and filter reordering strategies and evaluate them on Bench-U. (7) *UQE* does not release its code, so we reimplement its *Filter* and *Agg* operators, along with its logical optimizations, and run them on Bench-U. Since these two systems are closed-source, we have no choice but faithfully reproduce their core algorithms and configurations. Bench-U evaluates *system design choices* rather than official engineering optimizations, our results reflect the relative efficacy of these algorithmic designs rather than a definitive verdict on the original systems. For a comprehensive evaluation, we adapted and modified these systems to support our evaluation. We also list all other details (e.g., evaluation prompts) in Appendix A.

Evaluation Metrics. Following existing works [20, 33], we measure accuracy, cost, and latency w.r.t. all queries. For queries without aggregation (*Select*, *Filter*, *Join*), the outputs are relational tables, we follow [33] and report the average precision, recall, and F1 score. Given a query Q , the set of tuples returned by a method is $T(Q)$, and the ground truth is $GT(Q)$. For each element $t \in T(Q)$, we evaluate whether it can be matched to a corresponding tuple in $GT(Q)$. Since exact string matching fails on lexical variants (e.g., “Los Angeles Lakers” vs. “Lakers”), we employ an LLM-as-a-judge during evaluation to verify semantic equivalence, treating them as correct matches if they denote the same entity. Conversely, aggregation queries output numeric values. We classify the evaluation of these predicted values into two distinct categories based on the specific aggregation functions. For *MIN* and *MAX*, we require a precise exact match ($x = gt$). This is because, for instance, when a query seeks to identify the youngest player, the result must correspond uniquely to the specific person with the minimum age. For *SUM*, *AVG*, and *COUNT*, we note that such aggregations are inherently tolerant to small deviations: a predicted value that is close to the ground truth is often acceptable in practice. This is well aligned with prior works [3, 4, 11] in approximate query processing (AQP), where approximate answers with bounded errors are considered valid and meaningful. Motivated by this, we explicitly measure the deviation between predicted values and ground truth. We follow the previous work [13] to calculate an accuracy score derived from the bounded relative estimation error for each group: $1 - \min(1, \frac{|x - gt|}{\min(|x|, |gt|) + \epsilon})$. Here, x represents the predicted value for a specific group, gt represents the ground truth value, and ϵ (10^{-5}) is a minimal constant added to prevent division by zero. The final accuracy score for the query is then calculated by averaging the scores across all groups. All our metrics are normalized to the range $[0, 1]$, where higher values indicate better performance. This normalization ensures consistency in both visualization and comparison throughout our benchmark evaluation. For LLM costs, we report the average number of tokens (in thousands) per document per query by using the LLM API, which counts the token consumption w.r.t. all data modalities, e.g., text and images. For latency, we report the mean execution time in seconds per document per query.

Environment. We implemented all experiments in Python 3.7 on an Ubuntu Server with four Intel(R) Xeon(R) Gold 6148 2.40GHz CPUs with 80 cores, 2 NVIDIA 4090 GPUs, 1TB main memory and 6TB SSD. The same environment ensures a fair comparison over systems. We adopt GPT-4.1-mini as the default LLMs for API calls, and Qwen3-Embedding-0.6B as the default embedding model.

4.2 Overall Performance Comparison (RQ1)

As shown in table 3, across the majority of evaluated datasets (*Art*, *CSPaper*, *Legal* and *Finance*), *Palimpzest* and *LOTUS* consistently achieve the highest accuracy, reaching approximately 0.58 in *Art*, 0.56 in *CSPaper*, 0.52 in *Legal*, and 0.39 in *Finance*. Specifically, in short documents (*Art*), *Palimpzest* and *LOTUS* achieve the lowest cost (2k tokens per document) at the same time. Their costs are lower than those of systems employing chunking strategies (e.g., *ZenDB*). The reason is that when processing short documents, multiple target attributes frequently appear within the same text chunk. Repeatedly feeding this shared chunk to the LLMs for different

Method	Art			CSPaper			Player			Legal			Finance			Healthcare		
	Acc.	Cost	Latency	Acc.	Cost	Latency	Acc.	Cost	Latency	Acc.	Cost	Latency	Acc.	Cost	Latency	Acc.	Cost	Latency
Evaporate	0.08	-	0.16	0.23	-	2.40	0.18	-	0.30	0.14	-	0.25	0.07	-	1.50	0.01	-	0.52
Palimpzest	0.58	1.99	1.40	0.56	27.47	7.70	0.69	9.22	3.83	0.52	8.83	3.07	0.39	176.35	10.55	0.24	14.55	4.76
LOTUS	0.57	2.52	4.13	0.55	34.49	8.22	0.68	12.16	4.18	0.51	12.41	2.84	0.38	256.98	15.87	0.22	21.75	5.55
DocETL	0.45	6.60	14.79	0.45	41.41	63.15	0.70	54.23	76.96	0.51	93.48	151.32	0.29	501.08	761.88	0.21	148.83	176.01
ZenDB	0.44	2.73	1.66	0.45	18.37	32.60	0.65	6.66	2.77	0.44	15.31	14.33	0.35	40.76	4.14	0.22	14.91	2.10
QUEST	0.44	2.05	1.38	0.44	10.63	2.39	0.66	3.83	1.52	0.45	6.21	1.50	0.35	32.65	3.49	0.27	7.08	1.91
UQE	0.41	0.75	5.79	0.34	29.25	5.91	0.53	5.39	5.01	0.37	7.06	5.18	0.29	78.65	5.91	0.17	9.30	0.63

Table 3: Overall Performance across 6 datasets.

semantic operators ultimately consumes more tokens than directly processing the complete document a single time.

Insight I: Chunking (or data retrieval) does not help on short documents, as most relevant information is already contained within a small context. In such cases, applying retrieval or chunk-level processing introduces unnecessary overhead, increasing cost without improving accuracy.

However, as the length of documents increases, the chunks relevant to semantic operators become increasingly separated. Consequently, chunking-based systems begin to achieve significant cost savings. Within the CSPaper, Legal, and Finance datasets, although Palimpzest and LOTUS achieve highest accuracy, they incur costs that are five times higher than other systems (e.g., ZenDB and QUEST) that utilize text chunking and retrieval strategies.

Moreover, we observe that Palimpzest achieves higher accuracy and reduced cost compared to LOTUS within these datasets. This is because, when executing the filter operator, LOTUS requires the LLM to directly generate a boolean judgment based on the entire document context. In contrast, Palimpzest first extracts the target attribute and subsequently evaluates whether the extracted value satisfies the specified condition. Decomposing the execution pipeline into these two distinct steps provides the model with highly explicit instructions, thereby achieving more accurate results than direct logical execution.

Observation I: In the case of the filter operator, extracting the attribute first and evaluating the filter thereafter lead to more accurate results than directly executing it with LLMs. This is because decomposing the filter into the above two steps provides LLMs with more explicit instructions.

Palimpzest achieves lower operational costs than LOTUS, as Palimpzest implements specific logical optimization techniques. When evaluating multiple filters, Palimpzest prioritizes filters with low selectivity to discard irrelevant documents early in the execution pipeline. This filtering strategy substantially decreases the volume of operations required during subsequent processing stages. Therefore, if the user prioritizes accuracy, Palimpzest is the most suitable system, as it achieves the highest accuracy while maintaining relatively low costs.

Furthermore, QUEST and ZenDB achieve the lowest cost. They achieve substantial cost savings not only by partitioning documents but also by implementing logical optimizations. Furthermore, QUEST is more cost saving compared to ZenDB because it dynamically optimizes the execution order of filters for each individual document.

This dynamical strategy prevents the computational waste compared to ZenDB. Specifically, ZenDB generates a unified logical plan that evaluates filters based on the average token cost across the entire dataset. However, the exact size of chunks corresponding to a specific attribute varies significantly among different documents.

Observation II: Implementing dynamic logical optimization at document level is an effective strategy for minimizing the operational cost of current systems.

In general, within CSPaper, Legal and Finance datasets, there currently exists no optimal solution that simultaneously achieves superior accuracy, reduced costs, and lower latency. Existing systems either process the complete document context to maximize extraction accuracy at the expense of cost and latency, or they employ retrieval or chunking strategies to minimize cost consumption while inevitably sacrificing accuracy.

Insight II: All existing systems struggle on long documents due to missing an effective data retrieval mechanism. That is, scanning the document with an LLM preserves complete context but is prohibitively expensive, while the systems that use data retrieval to filter data tend to erroneously prune the text chunks that contain valuable information.

However, in Player and Healthcare, Palimpzest and LOTUS no longer achieve the highest accuracy. Within the Player dataset, almost all systems achieve high accuracy, reaching approximately 0.68. This is because attributes in this dataset can be easily extracted using independent text chunks, which eliminates the necessity to scan the entire document. Therefore, both full document processing and chunking-based systems achieve high accuracy. DocETL achieves the highest accuracy of 0.70, since it leverages multiple agents to process these chunks. This collaborative reasoning capability enhances the overall execution accuracy. However, this marginal accuracy gain comes at an unacceptable price. The token cost of DocETL reaches 54.23k tokens per document and its latency jumps to 76.96 seconds per document. This is because the system repeatedly feeds chunks into the model and exhaustively evaluates all possible execution plans. Therefore, QUEST and ZenDB are recommended under such circumstances, as they achieve high accuracy while maintaining the lowest costs. And QUEST is the most cost-efficient, since its document-level logical optimization can further reduce cost by dynamically adapting execution plans for each document.

Insight III: When retrieval successfully captures all relevant information, systems can achieve high accuracy. However, different execution plans lead to different costs. In such cases, logical optimization strategies can identify more cost-efficient plans and save cost. Furthermore, document-level logical optimization can further reduce cost by dynamically adapting execution plans (e.g., filter ordering) for each document, thus yielding the most cost-efficient execution plan.

Within Healthcare, a lot of irrelevant and noisy information frequently misleads the LLMs. Consequently, systems (Palimpzest and LOTUS) that supply the entire document directly to the models exhibit poor performance. And QUEST offers the highest accuracy and lowest cost. This advantage is due to its hierarchical two-stage retrieval strategy. First, the system utilizes lightweight summaries to entirely discard massive volumes of irrelevant documents. This document-level filtering substantially reduces unnecessary token consumption. Second, the system evaluates the remaining documents to process only on the relevant chunks. This chunk-level filtering process not only saves cost but also prevents noisy context from degrading the overall accuracy. However, this strategy relies heavily on the quality of retrieval, making the design of robust, highly effective retrieval mechanisms a critical direction for future research.

Insight IV: A good data retrieval strategy is the key to analyzing long and complex documents. In addition to reducing costs, it can even improve accuracy by preserving all relevant information while filtering out noisy or irrelevant content.

We additionally observe that Palimpzest and LOTUS significantly outperform other systems by more than 10% in accuracy within the Art and CSPaper datasets. This substantial performance gap exists because these datasets contain multimodal information. Palimpzest and LOTUS implement specialized semantic operators explicitly designed to process images. Furthermore, the CSPaper defines some cross modal attributes, such as best performing baseline, that require complex cross modal understanding by combining texts with corresponding tables or images. By providing the complete document directly to the models, Palimpzest and LOTUS preserve the cross modal structural integrity, ensuring that the critical associations between text, tables, and images remain complete, but at a high cost. Therefore, current systems lack advanced strategies for multi modal data chunking and corresponding retrieval. Developing such strategies could significantly advance cost optimization efforts in future works.

Insight V: Existing methods perform poorly on multi modal datasets, as they lack a retrieval method that can effectively prune the irrelevant data, while still preserve the correlation among the relevant data across different data modalities.

In general, we observe a direct correlation between operational cost and execution latency. This relationship exists because system latency fundamentally depends on the input & output tokens as well as the number of LLM calls. DocETL achieves the most time-consuming, as it applies the multi-agent technique that calls LLMs

multiple times, resulting in the highest overall token usage. LOTUS is the next because it calls LLMs multiple times and each time it feeds the entire document into LLMs. Palimpzest follows because it incorporates the chain-of-thought mechanism, leading to more outputs. UQE is relatively efficient, but still processes entire documents for each call. However, when filters are applied, its regression model significantly improves the efficiency. QUEST and ZenDB are efficient because they consume fewer input tokens and trigger fewer LLM calls. Evaporate is the most efficient because it generates code for extraction.

4.3 Overall Accuracy Comparison (RQ2)

Figure 3 presents the effectiveness results for Select queries. Evaporate performs the worst because it uses LLM-generated code for extraction. This code corresponds to a small set of rules, which are less effective for complex documents. Almost all systems perform well on dataset Player, with an F1 score around 0.85 because the dataset contains easy-to-extract attributes. DocETL performs the best because it leverages multi-agent techniques for extraction.

On datasets Legal, Finance, and Healthcare, with long documents and multiple challenging attributes, the accuracy varies across different systems. Specifically, LOTUS, UQE, and Palimpzest achieve similar performance and are top-ranked on the first three datasets, with an accuracy around 0.66 on Legal, and 0.52 on Finance, as they feed each entire document to LLMs and fully leverage the models’ in-context reasoning abilities. However, in Healthcare, a lot of irrelevant and noisy information can mislead LLMs. Besides, DocETL always selects the plan that splits documents into chunks, feeds each one to the LLM for extraction, produces multiple candidate answers per attribute and finally aggregates them. It is less effective on Healthcare and Finance because long documents introduce many noisy chunks, leading to incorrect extractions. For most datasets, QUEST and ZenDB perform worse because they only provide relevant chunks to LLMs to save cost. However, their performance is different on Healthcare, which lacks structured information. In this case, ZenDB feeds the entire document to LLMs, resulting in performance similar to that of LOTUS, UQE, and Palimpzest. However, since QUEST only retrieves and feeds LLMs the relevant chunks, it filters out much noisy information and achieves higher accuracy.

Summary I: For simple datasets with short content and easy-to-extract attributes, almost all systems perform well. For most datasets, the systems that feed entire documents to LLMs perform the best because retrieval-based strategies may miss in-context information. However, for datasets containing significant noise, systems supporting retrieval-then-extraction perform better due to filtering irrelevant and noisy context.

Overall, from the data perspective, the performance of all systems drop significantly on Finance and Healthcare, with F1-scores below 0.5 because : (1) noisy information: in Healthcare, individual documents frequently combine content from unrelated topics. For example, a single medical report might describe hypertension management in the first half and then switch to dermatitis treatment in the second. (2) Long context length: Finance documents are extremely long, making it difficult for an LLM to locate segments relevant to the target attribute within such massive texts. (3) Complex

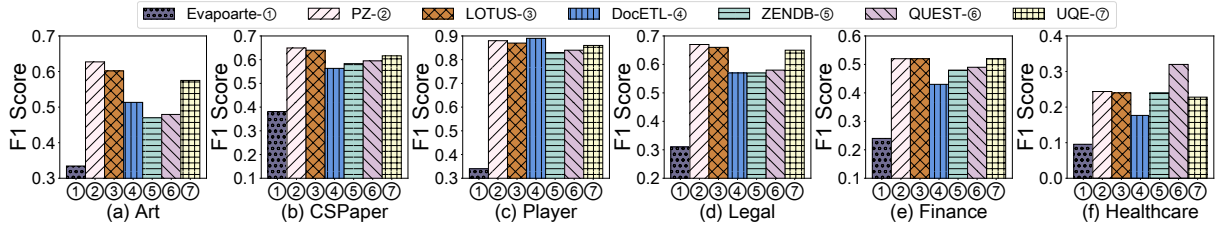


Figure 3: F1-score Comparison of Select Queries.

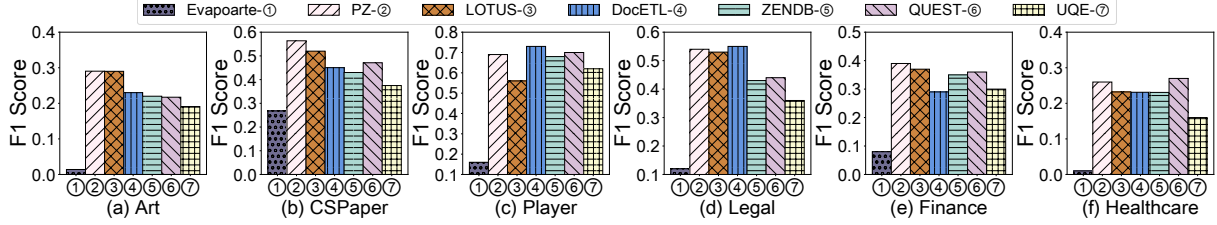


Figure 4: F1-score Comparison of Filter Queries.

attributes: Certain attributes require multi-step reasoning rather than direct extraction. For example, the attribute `Business_cost` in Finance is not explicitly stated but must be calculated by extracting and aggregating various cost components (e.g., raw materials cost and wages for car manufacturers). Existing systems, which most rely on a single LLM pass, fail to extract such attributes effectively.

Opportunity I: Extraction is the fundamental operator in UDA systems, but existing systems fall short in extracting complex attributes from complex and noisy datasets. This underscores the need for finer-grained retrieval, decomposition-based or multi-round extraction in the future.

On the Art and CSPaper datasets (Fig. 3), which heavily rely on visual information, LOTUS, UQE, and Palimpzest demonstrate superior performance due to their native support for multi-modal inputs. However, the performance variance among them is marginal, as all three systems rely solely on multi-modal LLMs to handle this information. No additional optimizations are conducted, such as cross-modal validation to verify consistency and joint reasoning to extract attributes from diverse modalities. Furthermore, existing systems require users to manually specify the modality of source data (text vs. image) for each attribute, instead of automatically mapping target attributes to the appropriate source data.

Opportunity II: This highlights the opportunity to develop a unified mechanism capable of automatically inferring attribute modalities, guiding them to suitable models, and leveraging cross-modal retrieval for more accurate information extraction.

Figure 4 shows the effectiveness of Filter queries. Existing systems implement filters in two ways. Given a filter, LOTUS and DocETL feed the description of the filter together with the document into an LLM and ask it to directly output a boolean output. Other systems first extract the attribute w.r.t. the filter and then determine whether the extracted value satisfies the filter. We observe that LOTUS performs worse than Palimpzest, ZenDB, QUEST on Player, as extracting the relevant information before evaluating the filter leads to more accurate answers. DocETL still achieves the best performance due to its multi-agent strategies. On complex datasets, LOTUS outperforms many systems because it feeds the entire document into LLMs, whereas the performance of DocETL declines due

to the imperfect chunking strategy. UQE performs worse than other systems because it samples a subset of documents to train a regression model, which is then used to predict whether the remaining documents satisfy the filter.

Summary II: For the filter operation, extracting the attribute first and evaluating the filter thereafter lead to more accurate results than directly executing it with LLMs. This is because decomposing the filter into the above two steps provides LLMs with more explicit instructions.

4.4 Overall Cost Comparison (RQ3)

Table 4 shows the cost of Select queries. On the dataset Art with short documents, systems that use chunking strategies (i.e., QUEST, ZenDB, and DocETL) consume more tokens than simply processing entire documents (UQE). The reason is that for each attribute, QUEST and ZenDB select attribute-related chunks. Given multiple attributes, there tends to be a number of duplicated chunks especially when the documents are short. As a result, the total token consumption exceeds that of simply processing full documents. The same pattern shows on images, which are more expensive to process than text documents. DocETL incurs the highest cost because (1) it examines all the chunks. When preprocessing each chunk, it also feeds adjacent chunks into LLMs; and (2) require executing all possible plans on sampled documents to select the optimal one. Palimpzest and LOTUS also process entire documents, but the chain-of-thought mechanism in Palimpzest leads to more output tokens, while LOTUS processes each attribute separately, multiplying the cost by the number of attributes.

Summary III: For datasets with short documents, strategies that feed the entire document to LLMs and extract attributes all at once are the most cost-effective.

On datasets CSPaper, Player, Legal, Finance and Healthcare with long documents, we can observe significant differences in cost across different systems. DocETL incurs the highest cost because of feeding a number of repeated chunks into the LLM and evaluating possible execution plans. LOTUS follows, as it repeatedly feeds the entire document into the LLM for multiple attributes. Palimpzest and UQE are next, as both feed the entire document to the LLM.

QUEST and ZenDB, which employ chunking strategies, cost less. In particular, QUEST is more cost-effective than ZenDB, as it feeds more fine-grained chunks into the LLM, further reducing the cost. Evaporate is the most cost-efficient strategy because it only uses LLMs to analyze several documents for code generation and then runs the code for extraction without additional LLM calls.

Summary IV: For datasets with long documents, strategies that retrieving attribute-related chunks instead of scanning entire documents are more cost-effective without sacrificing accuracy much.

On Healthcare, QUEST offers better cost-effectiveness than most baselines (excluding Evaporate). This advantage is due to its hierarchical two-stage retrieval strategy: (1) Document-level retrieval, which utilizes lightweight summaries to filter out massive amounts of irrelevant documents, and (2) Chunk-level retrieval, which further filters the irrelevant chunks within the remaining documents. This dual mechanism significantly reduces the tokens sent to LLMs. Moreover, QUEST attains an accuracy comparable to systems that exhaustively scan the entire corpus. Systems that scan the full corpus can indeed find the largest number of documents related to the target attributes. However, they typically do not filter input data at the chunk level, thus introducing substantial noise and harming accuracy. In contrast, although the document-level retrieval of QUEST may risk excluding some relevant documents, it prevents feeding much noise into the LLM. As a result, these effects offset each other, making QUEST reach an overall F1-score similar to exhaustive scanning approaches, but at a significantly lower cost. This indicates that the two-stage retrieval strategy is crucial: when retrieval is sufficiently accurate, this strategy can deliver considerable cost reductions without compromising accuracy.

Summary V: On large-scale datasets, employing a multi-stage retrieval strategy (filtering both documents and irrelevant chunks) is beneficial for cost reduction. However, this strategy relies heavily on the quality of retrieval, making the design of robust, highly effective retrieval mechanisms a critical direction for future research.

Table 5 shows the cost of Filter queries. UQE incurs a lower cost on most datasets because it trains a regression model to determine whether each filter condition is satisfied. Chunking-based systems like ZenDB and QUEST reduce token usage compared to Palimpzest, LOTUS, and DocETL by feeding only relevant chunks and using logical optimizations like prioritizing low selectivity and computational cost filters. Datasets Legal and Healthcare lack clear structures, leading to larger chunks in ZenDB and potential duplication when processing multiple attributes. Palimpzest is more cost-effective than LOTUS and DocETL because it employs a logical plan that prioritizes filters with low selectivity.

Summary VI: For queries with filters, QUEST and ZenDB apply logical optimization and chunking-based strategies to reduce costs while maintaining accuracy.

Overall, based on the above two sets of experiments, we observe that on complex datasets, all systems have a relatively low accuracy (e.g., 0.5-0.6 F1-score) and incur high cost. The reasons are two-fold. (1) The documents are long, including much noisy information that misleads LLMs, while the chunk strategies are not perfect. (2) Some of the attributes are difficult to extract (e.g., first_judge, which indicates whether a judgement was the first judgement), as it

might involve analyzing multiple chunks and employing inferential reasoning. This reveals research opportunity as below.

Opportunity III: A promising direction is to design a sophisticated chunking approach to produce chunks containing precise information w.r.t. the to-be-extracted attribute, leading to a high-quality and cost-effective strategy for UDA analysis over long documents.

4.5 Overall Latency Comparison (RQ4)

Table 4 and 5 compare the query latency of different systems. We observe that DocETL is the most time-consuming, as it applies the multi-agent technique that calls LLMs multiple times, resulting in the highest overall token usage. LOTUS is the next because it calls LLMs multiple times and each time it feeds the entire document into LLMs. Palimpzest follows because it incorporates the chain-of-thought mechanism, leading to more outputs. UQE is relatively efficient, but still processes entire documents for each call. However, when filters are applied, its regression model significantly improves the efficiency. QUEST and ZenDB are efficient because they consume fewer input tokens and trigger fewer LLM calls. Evaporate is the most efficient because it generates code for extraction.

Summary VII: Overall, latency is closely related to the input & output tokens as well as the number of LLM calls. Hence, chunking-based strategies with logical optimization (i.e., QUEST and ZenDB) are the most efficient because they reduce both the number of input tokens and LLMs calls.

4.6 Evaluation of Logical Optimization (RQ5)

Filter Reordering. We compare with four filter reordering strategies as follows. (1) Random: the filters are executed in random order; (2) Sel: the filters are ordered based on the selectivity; (3) Sel+Cost (ZenDB): the filters are ordered based on both the selectivity and the estimated average cost of extracting each attribute from the sampled documents; (4) (Sel+Cost)/doc (QUEST): each document has its own plan considering the selectivity and the estimated extraction cost w.r.t. this document. We compare the above strategies based on QUEST. To evaluate sufficiently, we conduct experiments on Player over different query subcategories w.r.t. different numbers of filters: S1 with one filter, S2 with 2 filters, and S3 with 3 or more. In Figure 5, for queries in S1, all baselines have nearly the same cost since each query uses a single filter and thus only one order. For queries with more filters, these methods are ranked as follows by the LLMs cost: (Sel+Cost)/doc < Sel+Cost < Sel < Random. The first two strategies save more cost because they optimize the order considering the cost. (Sel+Cost)/doc is the most cost-effective because it provides fine-grained optimization for each individual document.

Filter Pushdown. We compare with two strategies as follows. (1) DB-Pushdown: like in traditional databases, it pushes down filters respectively to the relevant tables before join. (2) Trans-Join (QUEST): it transforms a join to a filter operation and orders the filters using the above (Sel+Cost)/doc strategy to reduce the cost. We compare the above strategies using Mixed queries with filters on Player and Healthcare. We observe from Figure 6 that Trans-Join is more cost-effective than DB-Pushdown because given two tables to join, Trans-Join builds a cost model to judiciously determine which table will be extracted first and transformed to a filter on the other

Method	Art		CSPaper		Player		Legal		Finance		Healthcare	
	Cost	Latency	Cost	Latency	Cost	Latency	Cost	Latency	Cost	Latency	Cost	Latency
Evaporate	-	0.16	-	2.40	-	0.30	-	0.25	-	1.50	-	0.52
Palimpzest	1.64	1.34	20.08	5.82	6.67	1.43	6.96	1.39	138.90	8.38	10.80	2.36
LOTUS	2.33	1.47	32.37	6.31	12.41	1.75	9.93	1.43	297.85	13.24	25.10	3.28
DocETL	7.04	15.86	16.04	48.63	55.53	79.05	154.26	270.88	818.10	1509.08	184.30	304.96
ZenDB	1.94	1.18	4.98	3.61	3.72	0.98	3.92	0.94	32.33	3.55	10.31	1.92
QUEST	1.20	0.89	6.52	1.73	2.06	0.92	3.09	0.89	29.30	2.65	4.70	1.72
UQE	0.92	0.83	42.54	3.67	5.96	1.03	6.12	1.04	124.02	5.83	10.09	1.91

Table 4: Cost and Latency Comparison for Select Queries.

Method	Art		CSPaper		Player		Legal		Finance		Healthcare	
	Cost	Latency	Cost	Latency	Cost	Latency	Cost	Latency	Cost	Latency	Cost	Latency
Evaporate	-	0.16	-	2.40	-	0.30	-	0.25	-	1.50	-	0.52
Palimpzest	2.35	1.46	34.86	9.57	11.77	6.23	10.70	4.75	213.80	12.71	18.30	7.16
LOTUS	4.70	6.78	36.61	10.12	11.91	6.61	14.89	4.25	216.10	18.49	18.40	7.81
DocETL	6.15	13.71	26.77	77.66	52.92	74.86	32.70	31.75	184.06	14.67	113.35	47.06
ZenDB	3.51	2.13	71.76	61.59	9.59	4.56	26.69	27.72	49.18	4.73	19.50	2.28
QUEST	2.90	1.86	14.73	3.05	5.60	2.12	9.33	2.10	36.00	4.33	9.46	2.10
UQE	0.97	0.74	15.96	2.15	4.81	0.99	8.00	1.32	33.27	3.98	8.50	1.34

Table 5: Cost and Latency Comparison for Filter Queries.

table, which provides the opportunity to run a join first if it incurs a smaller data extraction cost.

Join order. We evaluate three strategies as follows. (1) Pushdown: all filters are pushed down first, and then join is performed; (2) Random: tables (document subsets) are joined in random order, with each pair of tables joined using Trans-Join; (3) Dynamic (ZenDB, QUEST): it uses a cost model to dynamically identify two tables to join in a left-deep manner, with each pair of tables joined using Trans-Join. We compare the above strategies using Join queries involving three tables on Art and Healthcare. We observe from Figure 7 that Dynamic saves much cost because it dynamically selects the join operation that leads to the lowest cost.

4.7 Evaluation of Physical Optimization (RQ6)

Model Selection. We evaluate two strategies in Palimpzest as follows. (1) Model-Sel: We define a set of candidate models (GPT-4.1-nano (Nano), GPT-4.1-mini (Mini), and GPT-4.1 (Normal)) and set the selection objective as “minimize cost while achieving the target accuracy.” (2) no-Model-Sel: The system always uses a specific model (Mini) for all queries. The user has to specify a desired accuracy. In our experiments, we set this value as the average accuracy obtained by running Filter queries with Mini on each dataset (Art and Legal). We evaluate both strategies using Filter queries on Art and Legal. The cost is calculated by multiplying the number of tokens by the model’s price per token. As shown in Figure 8, on Art, Model-Sel exceeds the target accuracy, while also reducing cost by using a cheaper model. This is achieved by assigning Normal to harder attributes and Nano to easier ones, while Art has more easy attribute than hard attributes. On the challenging Legal dataset, Model-Sel outperforms No-Model-Sel in accuracy, but with higher cost due to more frequently using Normal.

Model Cascades. We evaluate two strategies in LOTUS as follows. (1) Cascades: we construct a model cascade using two models, GPT-4.1-nano (Nano) and GPT-4.1-mini (Mini). For each query, the system first uses the lower-cost Nano to execute the queries. If the accuracy does not meet the desired accuracy, the query is then forwarded to the larger Mini for further processing. (2) No-Cascades:

The system uses only a single model, Mini to process all queries. We evaluate both strategies on the same set of queries for each dataset and use the same cost and target accuracy settings as the above model selection experiment. As shown in Figure 9, on Art, Cascades achieves the target accuracy while reducing cost by primarily using the less expensive Nano. This is because, for most attributes in Art, the cheaper model is sufficient. On the challenging Legal dataset, Cascades perform similar as No-Cascades, but incurs higher costs. This is because almost every attribute eventually requires Mini, and all documents have to be first processed by Nano before being fed to Mini.

Parallel Execution. We evaluate two strategies on Healthcare using Mixed queries as follows. (1) Parallel: employing parallel execution with different levels of parallelism, specifically with 2-way, 4-way, and 8-way thread parallelism. (2) no-Parallel: using a sequential approach without any parallelism. As shown in Figure 12, we observe that increasing the levels of parallelism leads to a proportional reduction in execution time. For example, with 2-way, 4-way, and 8-way thread parallelism, the execution time is reduced from 1.46 seconds (no-Parallel) to 0.74, 0.38, and 0.2 seconds, respectively, while the F1 score remains unchanged at about 0.53.

Opportunity IV: Currently, there is no system supporting end-to-end query optimization, including all above logical and physical optimization strategies, which would be a promising direction to achieve high-efficacy query execution.

Optimization for Join. We evaluate two join strategies on Healthcare. (1) Extract-All: it extracts the join key attribute from the two tables and join them. (2) with-Search: it extracts each value of the join key attribute from one table, leverages it as a semantic search key to prune many documents in the other table and then join. We can observe from Figure 11 that with-Search is more cost-effective than Extract-All since many documents in the other table are pruned without extraction. However, the accuracy is low because it incorrectly prunes documents that can be joined.

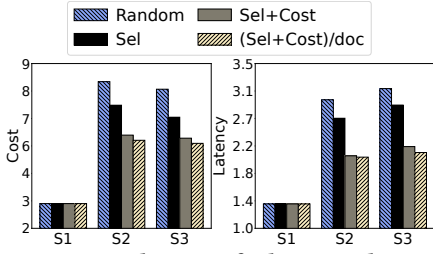


Figure 5: Evaluation of Filter Reordering.

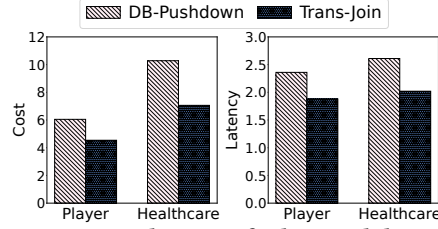


Figure 6: Evaluation of Filter Pushdown.

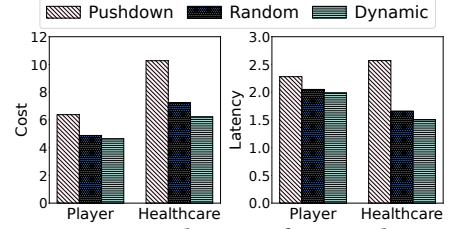


Figure 7: Evaluation of Join Order.

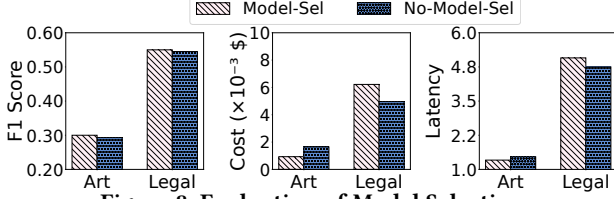


Figure 8: Evaluation of Model Selection.

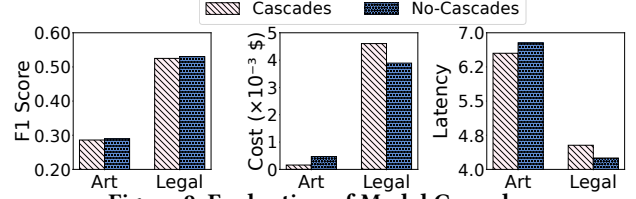


Figure 9: Evaluation of Model Cascades.

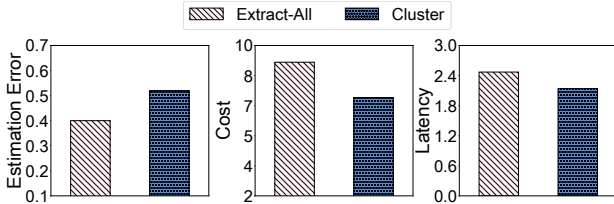


Figure 10: Aggregation Optimization.

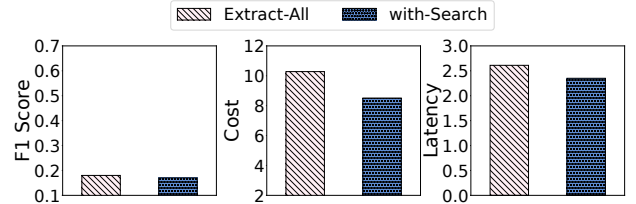


Figure 11: Join Optimization.

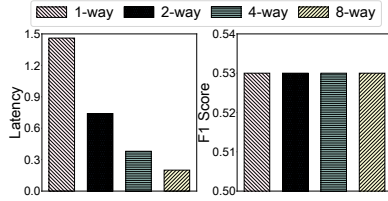


Figure 12: Evaluation of Parallel Execution.

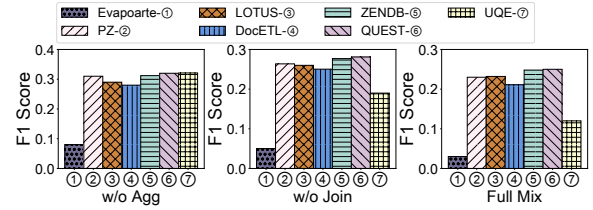


Figure 13: Average F1-score of Mixed Queries.

Optimization for Aggregation. We use the estimation error as the metric following [13], defined as the absolute difference between the estimated and true values divided by the true value. We evaluate two strategies on Art using Agg queries as follows. (1) Extract-All: it extracts the attributes involved in Groupby and Aggregation and then executes the query. (2) Cluster: it clusters the attribute-related chunks based on the attribute in Groupby, extracts the attribute in Aggregation within each cluster and executes the query. In Figure 10, Cluster is more cost-effective than Extract-All because it assumes all documents in a cluster share the same Groupby attribute value and thus avoids extracting that attribute. However, its relative error is high because high-quality clustering based solely on chunk embeddings is difficult.

Opportunity V: Another promising direction is to develop more effective strategies to align the embeddings of attributes with the embeddings of their chunks. In this way, the clustering in aggregation and pruning in join can be more accurate, and thereby the costs can be reduced more.

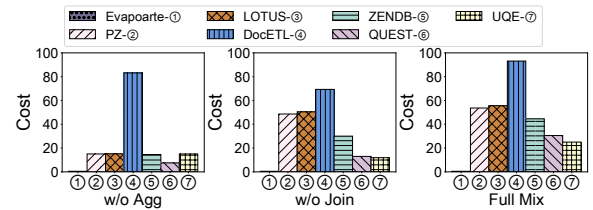


Figure 14: Average Cost of Mixed Queries.

4.8 Evaluation of Mixed Queries

We discuss the performance of different systems on three representative Mixed queries: Select +Filter +Join, Select +Filter +Agg, and Select +Filter +Agg +Join.

Select +Filter +Join.(w/o Agg) Systems like QUEST and ZenDB, which decompose the task into retrieval, attribute extraction, and then relational joins, consistently outperform methods that ask LLMs to perform semantic joins directly. This step-by-step process yields more reliable answers. However, QUEST's accuracy on these queries is on average 0.32 F1-score, 10% lower than on Filter

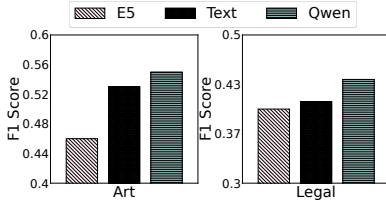


Figure 15: Ablation Study of Embedding Models.

queries because of error propagation: incorrect filters may drop relevant documents, and imperfect extraction of join keys further harms join performance. The cost difference between these two types of systems is modest, since both process multiple attributes and apply filters and joins. Still, QUEST and ZenDB significantly cut total cost by sending only retrieved chunks, rather than full documents, to the LLM.

Select + Filter + Agg.(w/o Join) Similarly, approaches that first extract structured tables and then run exact aggregations obtain the best accuracy with 0.28 F1-score. By decoupling extraction from aggregation, they avoid requiring the LLM to reason over large, noisy contexts. In contrast, semantic aggregation methods (LOTUS, DocETL) perform worse because the LLM must interpret all relevant text and compute aggregations (e.g., count, max) directly in the prompt. UQE has the lowest accuracy because it approximates aggregation via sampling instead of exact group-by, but it is also the cheapest since it aggregates only over samples and skips full extraction. QUEST and ZenDB remain cost-effective by using chunk-level extraction and performing relational group-by after extraction, avoiding multi-step LLM reasoning.

Select + Join + Agg + Filter.(Full Mix) All systems experience error propagation, causing the accuracy to fall to the level of the weakest mixed query. The cost remains high due to the execution many operators. Approaches that pass entire documents to LLMs (LOTUS, Palimpzest) incur the greatest expense, whereas QUEST and ZenDB save cost by using retrieval-based mechanism.

Opportunity VI: A promising research direction is to design an end-to-end optimization framework that simultaneously consider retrieval, extraction, filtering, joining, and aggregation. This would significantly enhance accuracy and cost-efficiency in complex analytical settings.

4.9 Ablation Studies

Ablation Study of Embedding Models. We evaluate different embedding models to analyze whether using a better embedding model can improve the system performance. We utilize multilingual-e5-large (E5) [38], text-embedding-3 (Text) [25] and Qwen3-Embedding (Qwen) [40]. On MTEB benchmark [23], their performance is ranked as follows: Qwen > Text > E5. We evaluate the performance of different embedding models by replacing the embedding model in QUEST and testing on the Art, and Legal datasets, each with Filter queries. As shown in Figure 15, system performance improves as the quality of the embedding model increases. This is because better embedding models provide more accurate semantic representations, resulting in more precise chunk retrieval, and thus improve overall performance.

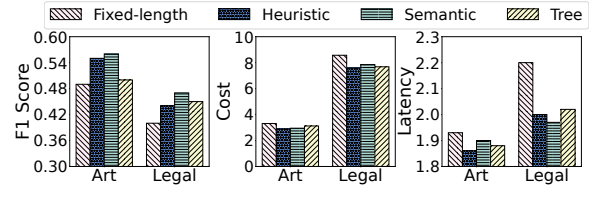


Figure 16: Ablation Study of Chunking Strategies.

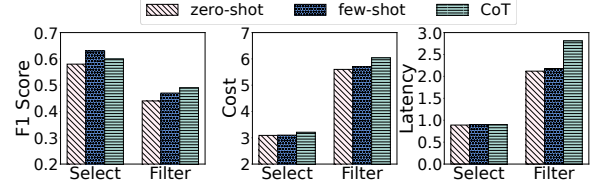


Figure 17: Ablation Study of Prompts.

Ablation Study of Chunking Strategies. We evaluate how different chunking strategies affect system performance: (1) Fixed-length [9]: fixed 512-token chunks. (2) Heuristic [31]: grammar-based chunks, up to 512 tokens. (3) Semantic (QUEST): semantic chunks, 128–512 tokens. (4) Tree (ZenDB): SHT tree-based splitting. We run this experiment by changing the chunking strategy in QUEST while running Filter queries on Art and Legal. Semantic performs best by capturing rich semantic representations, yielding superior embedding similarity. Tree surpasses Heuristic on Art and Legal because it preserves more tokens per chunk, providing richer context at the cost of higher latency and expense. Fixed-length performs worst, as it can split sentences across chunks, leading to incomplete information.

Ablation Study of Prompts. We evaluate the impact of different prompt design strategies on system performance to determine whether prompt selection will lead to improved results. The prompt design strategies we consider are as follows: (1) Zero-shot: directly provides the task instruction to the model without any examples. (2) Few-shot: provides the task instruction along with a few annotated examples. (3) CoT (Chain-of-Thought): incorporates explicit reasoning steps or prompts the model to generate intermediate reasoning, enabling multi-step problem solving. We evaluate the performance of different prompt design strategies by replacing the prompts in QUEST and testing with Select and Filter queries on the Legal dataset. As shown in Figure 17, for extraction tasks, all three approaches perform similarly, with Few-shot slightly outperforming others due to its detailed examples, and CoT outperforming Zero-shot due to its guided reasoning. For queries with filters, CoT achieves the best performance by guiding the model through step-by-step reasoning. However, CoT produces longer output sequences, resulting in higher latency and greater token costs compared to the other two approaches. Additionally, Few-shot requires more input tokens than Zero-shot. Few-shot ranks next, as concrete examples help the model learn the task pattern. Zero-shot performs worst, as it lacks examples and reasoning guidance.

Ablation Study of Different LLMs. We evaluate the impact of different types of LLMs, including close-source models (GPT-4.1 (GPT), Gemini-2.5-pro (Gemini) and Claude-sonnet-4 (Claude)), and open-source models (Deepseek-V3 (Deepseek),

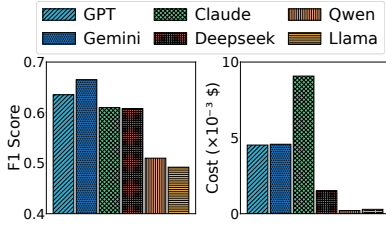


Figure 18: Ablation Study of Different LLMs.

Qwen3-30B-A3B (Qwen) and Llama-4-scout-17b (Llama)) on system performance to assess whether stronger LLMs lead to improved results. To ensure fair comparison, we use the same third-party LLM service provider (OpenRouter[2]) and invoke all models via API. We then evaluate the performance of various LLMs by replacing the LLM in QUEST and testing each model on Select queries from the Legal dataset. Figure 18 presents the results, with performance ranked as: Gemini > GPT > Claude > Deepseek > Qwen > Llama, while the inference cost per document is ranked as Qwen < Llama < Deepseek < GPT < Gemini < Claude. We find that higher inference costs do not necessarily guarantee superior performance, and the gap between open-source and close-source models is narrowing rapidly.

Scope and Limitations. First, Bench-U evaluates relational-style declarative querying over unstructured data, targeting systems (e.g., Palimpzest, LOTUS, UQE) that expose such data as relational views. While enabling rigorous evaluation of operators (Select, Filter, Join, Agg) and optimizers, it underrepresents non-tabular tasks such as open-ended QA or schema-free summarization. Second, results depend on design choices such as LLMs and prompting. Changing these may alter performance. Thus, Bench-U is not intended to provide a universal ranking, but a unified platform for comparing designs under strict settings, helping researchers uncover common observations and promising architectural directions within specific constraints.

5 Related Work

UDA Benchmark. Recent years have witnessed substantial progress in benchmarking and structuring unstructured data. Pioneering works such as Doctopus and Evaporate focus on benchmarking the extraction of structured attributes from unstructured data to construct tables. SemRank focuses on benchmarking the retrieval and prioritization of relevant documents for a given query. Bench-U instead targets executing relational-style queries directly over unstructured data, moving beyond the simple extract operator and retrieval strategies. Other existing systems evaluate few queries to highlight specific capabilities, and thus provide limited coverage of the overall design space. For example, Palimpzest demonstrates cost optimization on a fraudulent email detection task over 1000 short emails, highlighting its ability to aggressively push down cheap filters. Similarly, LOTUS evaluates its execution engine using five queries across five tasks, focusing on semantic operators. For example, it tests the `sem_filter` operator via fact verification over 500 FEVER claims. However, the evaluations for LOTUS and Palimpzest do not expose the system challenges regarding text chunking. Furthermore, DocETL evaluates its architecture using one representative query per dataset across five datasets. For example, it runs a clause extraction task on CUAD (510 long legal

contracts), extracting text spans for predefined clause types. These tasks highlight multi-agent reasoning, but do not consider token cost optimization. To address these limitations, Bench-U provides a diverse query workload that enables systematic evaluation of optimization strategies. Concurrent with our effort, SemBench has a distinct and valuable goal: benchmarking semantic query processing over a mix of structured and unstructured data. However, unlike Bench-U, SemBench does not target thoroughly evaluating the capability of the LLM-powered data systems in processing long, complex unstructured documents that raise significant challenges in cost and accuracy.

LLM-powered Unstructured Data Analysis Systems. Lotus [27] introduces semantic operators for unstructured data processing, including indexing, extraction, filtering and joining capabilities that enable the construction of complex analytical pipelines. It provides optimized physical implementation for each operator but lacks logical optimizations. Palimpzest [21] offers libraries for users to write declarative Python code to analyze unstructured data. It optimizes the logical plan of each program via filter reordering based on selectivities. DocETL[32] focuses on improving the accuracy of UDA using a multi-agent strategy, but it does not conduct logical optimization to save cost. ZenDB[20] uses semantic hierarchical trees to identify relevant document sections and applies filter ordering and predicate pushdown for optimization. However, it requires well-structured documents and is evaluated on just 221 documents and 27 queries. UQE [13] provides SQL-like analysis with sampling-based aggregation, while CAESURA [36] decomposes queries into operators to handle data in different modalities.

In addition, extracting data from unstructured sources is a key component in analyzing unstructured information, the strategies have evolved from rule-based methods [19, 26, 29, 30] to modern deep learning approaches [17, 39]. Nowadays, LLM-based systems seek to resolve these issues. Evaporate [5] generates extraction code through LLMs, balancing cost and quality through weak supervision. Similarly, early systems [12, 15, 34, 35] employ LLMs for document analysis but overlook cost optimization, which is a critical concern given the computational expense of LLM inference, and lack a thorough benchmarking.

6 Conclusion

In this paper, we build a comprehensive LLM-powered benchmark for unstructured data analysis. We collect six diverse unstructured datasets, define key attributes, and label them manually with the assistance of LLMs in cross-validation. We then construct hundreds of queries with various analytical operators, capable of comprehensively assessing the capability of existing systems, run the queries on the labeled datasets, and analyze their performance in depth.

References

- [1] [n.d.]. arXiv. <https://arxiv.org>. Accessed: 2025-02-17.
- [2] [n.d.]. OpenRouter: Unified API access to multiple large language models. <https://openrouter.ai/>. Accessed: 2025-11-17.
- [3] Swarup Acharya, Phillip B. Gibbons, Viswanath Poosala, and Sridhar Ramaswamy. 1999. The Aqua approximate query answering system. In *Proceedings of the 1999 ACM SIGMOD International Conference on Management of Data* (Philadelphia, Pennsylvania, USA) (SIGMOD '99). Association for Computing Machinery, New York, NY, USA, 574–576. <https://doi.org/10.1145/304182.304581>
- [4] Sameer Agarwal, Aurojit Panda, Barzan Mozafari, Samuel Madden, and Ion Stoica. 2012. BlinkDB: Queries with Bounded Errors and Bounded Response Times on Very Large Data. arXiv:1203.5485 [cs.DB] <https://arxiv.org/abs/1203.5485>
- [5] Simran Arora, Brandon Yang, Sabri Eyuboglu, Avani Narayan, Andrew Hojel, Immanuel Trummer, and Christopher Ré. 2025. Language Models Enable Simple Systems for Generating Structured Views of Heterogeneous Data Lakes. arXiv:2304.09433 [cs.CL] <https://arxiv.org/abs/2304.09433>
- [6] Ron Artstein and Massimo Poesio. 2008. Inter-coder agreement for computational linguistics. *Comput. Linguist.* 34, 4 (Dec. 2008), 555–596. <https://doi.org/10.1162/coli.07-034-R2>
- [7] Art-.org. 2025. *Art-.org – Artists and Artworks (19th–21stC.)*. <https://www.art-.org/>
- [8] Australasian Legal Information Institute (AustLII). [n.d.]. AustLII – Australasian Legal Information Institute. <https://www.austlii.edu.au/>
- [9] Sinchana Ramakanth Bhat, Max Rudat, Jannis Spiekermann, and Nicolas Flores-Herr. 2025. Rethinking Chunk Size For Long-Document Retrieval: A Multi-Dataset Analysis. arXiv:2505.21700 [cs.IR] <https://arxiv.org/abs/2505.21700>
- [10] Chengliang Chai, Jiajun Li, Yuhao Deng, Yuanhao Zhong, Ye Yuan, Guoren Wang, and Lei Cao. 2025. Doctopus: Budget-Aware Structural Table Extraction from Unstructured Documents. *Proc. VLDB Endow.* 18, 11 (July 2025), 3695–3707. <https://doi.org/10.14778/3749646.3749647>
- [11] Surajit Chaudhuri, Bolin Ding, and Srikanth Kandula. 2017. Approximate Query Processing: No Silver Bullet. In *Proceedings of the 2017 ACM International Conference on Management of Data* (Chicago, Illinois, USA) (SIGMOD '17). Association for Computing Machinery, New York, NY, USA, 511–519. <https://doi.org/10.1145/3035918.3056097>
- [12] Zui Chen, Zihui Gu, Lei Cao, Ju Fan, Sam Madden, and Nan Tang. 2023. Symphony: Towards Natural Language Query Answering over Multi-modal Data Lakes. <https://www.cidrdb.org/cidr2023/papers/p51-chen.pdf>
- [13] Hanjun Dai, Bethany Yixin Wang, Xingchen Wan, Bo Dai, Sherry Yang, Azade Nova, Pengcheng Yin, Phitchaya Mangpo Phothilimthana, Charles Sutton, and Dale Schuurmans. 2024. UQE: A Query Engine for Unstructured Databases. arXiv:2407.09522 [cs.DB] <https://arxiv.org/abs/2407.09522>
- [14] Enterprise RAG Challenge. [n.d.]. Enterprise RAG Challenge. <https://rag.abdullin.com/>. Accessed 17 July 2025.
- [15] Saehan Jo and Immanuel Trummer. 2024. ThalamusDB: Approximate Query Processing on Multi-Modal Data. , 26 pages. <https://dl.acm.org/doi/10.1145/3654989>
- [16] K. V. Kanimozhi and M. Venkatesan. 2015. Unstructured Data Analysis—A Survey. *International Journal of Advanced Research in Computer and Communication Engineering* 4, 3 (2015), 223–225.
- [17] Keshav Kolluru, Vaibhav Adlakha, Samartha Aggarwal, Mausam, and Soumen Chakrabarti. 2020. OpenIE6: Iterative Grid Labeling and Coordination Analysis for Open Information Extraction. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, Online, 3748–3761. <https://doi.org/10.18653/v1/2020.emnlp-main.306>
- [18] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody-Hao Yu, JosephE Gonzalez, Hao Zhang, and Ion Stoica. [n.d.]. Efficient Memory Management for Large Language Model Serving with PagedAttention. ([n. d.]).
- [19] Taesung Lee, Zhongyuan Wang, Haixun Wang, and Seung-won Hwang. 2013. Attribute extraction and scoring: A probabilistic approach. In *2013 IEEE 29th International Conference on Data Engineering (ICDE)*. 194–205. <https://doi.org/10.1109/ICDE.2013.6544825>
- [20] Yiming Lin, Madelon Hulsebos, Ruiying Ma, Shreya Shankar, Sepanta Zeigham, Aditya G. Parameswaran, and Eugene Wu. 2024. Towards Accurate and Efficient Document Analytics with Large Language Models. arXiv:2405.04674 [cs.DB] <https://arxiv.org/abs/2405.04674>
- [21] Chunwei Liu, Matthew Russo, Michael Cafarella, Lei Cao, Peter Baille Chen, Zui Chen, Michael Franklin, Tim Kraska, Samuel Madden, and Gerardo Vitagliano. 2024. A Declarative System for Optimizing AI Workloads. arXiv:2405.14696 [cs.CL] <https://arxiv.org/abs/2405.14696>
- [22] Edward Loper and Steven Bird. 2002. NLTK: The Natural Language Toolkit. arXiv:cs/0205028 [cs.CL] <https://arxiv.org/abs/cs/0205028>
- [23] Niklas Muennighoff, Nouamane Tazi, Loïc Magne, and Nils Reimers. 2023. MTEB: Massive Text Embedding Benchmark. arXiv:2210.07316 [cs.CL] <https://arxiv.org/abs/2210.07316>
- [24] National Basketball Association – Wikipedia. [n.d.]. NBA – Wikipedia. https://en.wikipedia.org/wiki/National_Basketball_Association. Accessed 17 July 2025.
- [25] Arvind Neelakantan, Tao Xu, Raul Puri, Alec Radford, Jesse Michael Han, Jerry Tworek, Qiming Yuan, Nikolas Tezak, Jong Wook Kim, Chris Hallacy, Johannes Heidecke, Pranav Shyam, Boris Power, Tyna Eloundou Nekoul, Girish Sastry, Gretchen Krueger, David Schnurr, Felipe Petroski Such, Kenny Hsu, Madeleine Thompson, Tabarak Khan, Toki Sherbakov, Joanne Jang, Peter Welinder, and Lilian Weng. 2022. Text and Code Embeddings by Contrastive Pre-Training. arXiv:2201.10005 [cs.CL] <https://arxiv.org/abs/2201.10005>
- [26] Christina Niklaus, Matthias Cetto, André Freitas, and Siegfried Handschuh. 2018. A Survey on Open Information Extraction. arXiv:1806.05599 [cs.CL] <https://arxiv.org/abs/1806.05599>
- [27] Liana Patel, Siddharth Jha, Melissa Pan, Harshit Gupta, Parth Asawa, Carlos Guestrin, and Matei Zaharia. 2025. Semantic Operators: A Declarative Model for Rich, AI-based Data Processing. arXiv:2407.11418 [cs.DB] <https://arxiv.org/abs/2407.11418>
- [28] Pengcheng Qiu, Chaoyi Wu, Xiaoman Zhang, Weixiong Lin, Haicheng Wang, Ya Zhang, Yanfeng Wang, and Weidi Xie. 2024. Towards Building Multilingual Language Model for Medicine. arXiv:2402.13963 [cs.CL] <https://arxiv.org/abs/2402.13963>
- [29] Swarnadeep Saha and Mausam. 2018. Open Information Extraction from Conjunctive Sentences. , 2288–2299 pages. <https://aclanthology.org/C18-1194/>
- [30] Swarnadeep Saha, Harinder Pal, and Mausam. 2017. Bootstrapping for Numerical Open IE. , 317–323 pages. <https://doi.org/10.18653/v1/P17-2050>
- [31] Roie Schwaber-Cohen and Arjun Patel. 2025. Chunking Strategies for LLM Applications. Pinecone Blog. <https://www.pinecone.io/learn/chunking-strategies-for-llm-applications/> Retrieved 17 July 2025 from Pinecone website.
- [32] Shreya Shankar, Tristan Chambers, Tarak Shah, Aditya G. Parameswaran, and Eugene Wu. 2025. DocETL: Agentic Query Rewriting and Evaluation for Complex Document Processing. arXiv:2410.12189 [cs.DB] <https://arxiv.org/abs/2410.12189>
- [33] Zhaoze Sun, Qiyang Deng, Chengliang Chai, Kaisen Jin, Xinyu Guo, Han Han, Ye Yuan, Guoren Wang, and Lei Cao. 2025. QUEST: Query Optimization in Unstructured Document Analysis. arXiv:2507.06515 [cs.DB] <https://arxiv.org/abs/2507.06515>
- [34] James Thorne, Majid Yazdani, Marzieh Saeidi, Fabrizio Silvestri, Sebastian Riedel, and Alon Halevy. 2021. From Natural Language Processing to Neural Databases. , 1033–1039 pages. <https://doi.org/10.14778/3447689.3447706>
- [35] Matthias Urban and Carsten Binnig. 2023. Towards Multi-Modal DBMSs for Seamless Querying of Texts and Tables. arXiv:2304.13559 [cs.DB] <https://arxiv.org/abs/2304.13559>
- [36] Matthias Urban and Carsten Binnig. 2024. CAESURA: Language Models as Multi-Modal Query Planners. In *14th Conference on Innovative Data Systems Research, CIDR 2024, Chaminade, CA, USA, January 14-17, 2024*. www.cidrdb.org. <https://www.cidrdb.org/cidr2024/papers/p14-urban.pdf>
- [37] Bin Wang, Chao Xu, Xiaomeng Zhao, Linke Ouyang, Fan Wu, Zhiyuan Zhao, Rui Xu, Kaiwen Liu, Yuan Qu, Fukai Shang, Bo Zhang, Liqun Wei, Zhihao Sui, Wei Li, Botian Shi, Yu Qiao, Dahua Lin, and Conghui He. 2024. MinerU: An Open-Source Solution for Precise Document Content Extraction. arXiv:2409.18839 [cs.CV] <https://arxiv.org/abs/2409.18839>
- [38] Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024. Multilingual E5 Text Embeddings: A Technical Report. arXiv:2402.05672 [cs.CL] <https://arxiv.org/abs/2402.05672>
- [39] Bowen Yu, Yucheng Wang, Tingwen Liu, Hongsong Zhu, Limin Sun, and Bin Wang. 2021. Maximal Clique Based Non-Autoregressive Open Information Extraction. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih (Eds.). Association for Computational Linguistics, Online and Punta Cana, Dominican Republic, 9696–9706. <https://doi.org/10.18653/v1/2021.emnlp-main.764>
- [40] Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. 2025. Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models. arXiv preprint arXiv:2506.05176 (2025).

A Evaluation

A.1 System Adaptation for Evaluation

We have made several modifications to the open source codes for different systems to ensure that all systems can operate within a unified environmental configuration, thus enabling the convenient acquisition of the test results and reliable conclusions. The specific modifications are detailed as follows.

Evaporate. The original code applied an exact matching approach when identifying attribute-related texts; however, the specified text may not necessarily appear in actual documents. To more accurately capture the content potentially requiring extraction, we have introduced a synonym matching function, enabling the system to recognize and extract synonymous texts relevant to the queried attributes.

(1) Synonym Matching

Original:

```
for chunk in chunks:
    if attribute.lower() in chunk.lower():
        cleaned_chunks.append(chunk)
```

New:

```
def match_with_synonyms(attribute, chunk):
    synonyms = ATTRIBUTE_SYNONYMS.get(attribute.lower(), [attribute])
    for syn in synonyms:
        if syn.startswith(r"\b"):
            if re.search(syn, chunk):
                return True
        else:
            if syn.lower() in chunk.lower():
                return True
    return False

for chunk in chunks:
    if match_with_synonyms(attribute, chunk):
        cleaned_chunks.append(chunk)
```

Palimpzest. During the testing process, we encountered several errors related to numerical and null-value attributes that we have resolved through targeted modifications. In addition, we have adjusted the method for invoking the LLM to ensure compatibility with the API interface utilized in our testing procedures.

(1) Empty Value

Original:

```
field_lengths = [(field, len(value) if value is not
                  None else 0) for field, value in context.items()]
```

New:

```
def _get_value_length(value):
    if value is None:
```

```
        return 0
    if isinstance(value, str):
        return len(value)
    if isinstance(value, (list, dict)):
        return len(str(value))
    # change to str and get the length
    return len(str(value))
field_lengths = [(field, _get_value_length(value))
                 for field, value in context.items()]
```

(2) Numeric Value

Original:

```
num_outputs = sum(record.passed_operator for record
                  in record_set.data_records)
```

New:

```
num_outputs = sum(record.passed_operator for record
                  in record_set.data_records if record.
                  passed_operator is not None)
```

(3) LLM

Original:

```
OpenAI(api_key=api_key)
```

New:

```
OpenAI(api_key=api_key, base_url=api_base)
```

LOTUS. To enable the model cascade function, we made simple modifications to this component.

(1) Synonym Matching

Original:

```
def _get_top_choice_logprobs(self, response:
                             ModelResponse) -> list[
    ChatCompletionTokenLogprob]:
    choice = response.choices[0]
    assert isinstance(choice, Choices)
    logprobs = choice.logprobs["content"]
    return [ChatCompletionTokenLogprob(**logprob)
            for logprob in logprobs]
```

New:

```
def _get_top_choice_logprobs(self, response:
                             ModelResponse) -> list[
    ChatCompletionTokenLogprob]:
    choice = response.choices[0]
    assert isinstance(choice, Choices)
    logprobs = choice.logprobs["content"]
    return logprobs
```

DocETL. We did not make any special modifications to this system.

A.2 Dataset Details

This section will first provide an overview of the dataset, then introduce all the attributes in each dataset in our benchmark and all detailed descriptions about them.

- **Art Dataset.** Art includes 20 textual attributes and 6 image attributes. These 26 attributes are combined into one table in the ground truth.
 - Art Text: Biographies and information about the artists.
 - Art Image: Visual features and characteristics of the images.
- **CSPaper Dataset.** CSPaper includes 19 attributes, of which 4 are mixed attributes, for example, some papers describe the performance of all baselines in the main text, while other papers only describe the best-performing baselines and leave other results in tables or figures, resulting in an analysis scenario with mixed-modal. These 19 attributes are combined into one table in the ground truth.
- **Player Dataset:** Player includes statistics of professional basketball players, teams, owners, and cities
 - Player: Personal information, draft history, and career achievements
 - Owner: Ownership details and personal demographics
 - Team: Organizational information and championship records
 - City: Geographic and economic data of NBA team cities
- **Legal Dataset:** Legal includes documentation of court cases and legal proceedings
- **Healthcare Dataset:**
 - Disease: Medical conditions and disease information
 - Drug: Pharmaceutical drugs and medication information
 - Medical Institutes: Medical research institutions and organizations
- **Finance Dataset:** Finance The data set consists of financial reports and company information.

(1) Art Text Dataset Attributes

This dataset contains 20 attributes capturing comprehensive biographical and career information about artists, including their personal details, artistic movements, educational background, and professional achievements.

Attribute	Description
name	Full name of the artist
nationality	Artist's nationality (e.g., french, japanese)
art_movement	Art movement contributed to (e.g., impressionism)
birth_date	Date in %Y/%-m/%-d format (e.g., 1839/1/9), or leave empty if not applicable
death_date	Date in %Y/%-m/%-d format (e.g., 1836/1/1), or leave empty if not applicable
age	Age of the artist (if death_date is given, compute as death_date - birth_date; otherwise, compute as 2025/1/1 - birth_date)
century	Century when the artist was active or influential (e.g., 19th or 19th–20th; can span multiple centuries)
zodiac	Western zodiac sign based on birth date, choose one form [Aries, Taurus, Gemini, Cancer, Leo, Virgo, Libra, Scorpio, Sagittarius, Capricorn, Aquarius, Pisces], separated by ' ' (e.g. Aries)

Attribute	Description
birth_country	Country where the artist was born (e.g., netherlands)
birth_city	City where the artist was born
birth_continent	Continent where the artist was born (e.g., europe, asia)
death_country	Country where the artist died, or leave empty if not applicable (e.g., switzerland)
death_city	City where the artist died, or leave empty if not applicable
field	Primary artistic field, choose one or more from: [Painting, Sculpture, Printmaking, Drawing, Photography, Illustration, Collage, Textile Art, Installation, Performance, Ceramics, Digital Art, Architecture, Mosaic, Graphic Design, Calligraphy], separated by ' ' (e.g. Collage Illustration Textile Art)
genre	Primary genre of work, choose one or more from: [Portrait, Landscape, Still Life, Religious, Mythological, History, Genre, Nude, Animal, Illustration, Allegorical, Abstract, Fantasy, Sketch, Figurative, Conceptual, Surrealist, Symbolic, Political, Social Realism, Pop Art, Narrative, Environmental, Sculpture, Architecture, Music, Film, Digital Art, Literature, Other], separated by ' ' (e.g. Portrait Political)
marriage	Marital status of the artist, reflecting the artist's latest (final) marital status as mentioned in the biography. Choose one from: [Cohabiting, Divorced, Married, Remarried, Separated, Unmarried, Widowed]. Only extract and fill in the last marital status referenced in the text
art_institution	Art academies, schools, organizations, associations, or exhibition societies the artist was affiliated with, taught at, or participated in (e.g., national art academies, artist unions, art schools, exhibition societies)
teaching	Whether the artist taught at an institution (1 if yes, 0 if none)
awards	Number of significant and clearly documented awards the artist received (enter an integer; use 0 if none; exclude medals, scholarships, or honorary mentions)

(2) Art Image Dataset Attributes

This dataset contains 6 attributes focusing on the visual characteristics and artistic elements of artworks, including style, composition, and color properties.

Attribute	Description
style	The style of the artwork (e.g., realism, abstract)
image_genre	The genre of the artwork (e.g., landscape, portrait)
object	The primary objects or subjects in the artwork
color	The predominant color used in the artwork
tone	The tonal qualities (e.g., bright, dark, neutral)
composition	The composition style of the artwork

(3) Player Dataset - Player Attributes

This dataset contains 13 attributes covering NBA players' personal information, draft history, career achievements, and championship records across different basketball competitions.

Attribute	Description
name	Full name of the NBA player (e.g., LeBron James)
birth_date	Birth date of the player; use format YYYY/%-m/%-d (e.g., 1984/2/30)
nationality	Nationality of the player (e.g., USA, France)
age	Age of the player (if death_date is given, compute as death_date - birth_date; otherwise, compute as 2025/1/1 - birth_date)
team	The current NBA team the player belongs to, or the last NBA team the player joined if not currently active (e.g., Los Angeles Lakers)
position	Player's primary position on the team; choose one from ['Frontcourt', 'Backcourt']
draft_pick	The pick number at which the player was selected in the NBA draft (e.g., 1). If the player was selected in the second round, add 30 to the pick number (e.g., 2nd round, 1st pick → 31)
draft_year	The year the player was selected in the NBA draft; use format YYYY (e.g., 2003)
college	The college the player attended, if applicable (e.g., Duke University)
nba_championships	Number of NBA championships the player has won (e.g., 4)
mvp_awards	Number of NBA MVP awards the player has won; use 0 if none (e.g., 2)
olympic_gold_medals	Number of Olympic gold medals the player has won (e.g., 3)
fiba_world_cup	Number of FIBA World Cup titles the player has won (e.g., 1)

(4) Player Dataset - Team Attributes *This dataset contains 5 attributes covering NBA teams' organizational information, historical data, geographic location, and championship achievements.*

Attribute	Description
team_name	Official name of the NBA team (e.g., Los Angeles Lakers, Boston Celtics)
founded_year	Year when the team was founded or established; use format YYYY (e.g., 1947)
location	Current city where the team is based and plays home games (e.g., Los Angeles, Boston)
ownership	Full name of the current team owner or primary owner (e.g., Jeanie Buss)
championships	Total number of NBA championships won by the team throughout its history (e.g., 17)

(5) Player Dataset - Owner Attributes *This dataset contains 5 attributes covering NBA team owners' personal information, ownership details, and tenure with their respective teams.*

Attribute	Description
name	Full name of the NBA team owner (e.g., Steve Ballmer, Jeanie Buss)
age	Current age of the team owner (e.g., 68)
nationality	Nationality of the team owner (e.g., USA, Canada)
NBA_team	Name of the NBA team owned by this individual (e.g., Los Angeles Clippers)
own_year	Year when the owner acquired or started owning the NBA team; use format YYYY-MM-DD (e.g., 2014-05-29)

(6) Player Dataset - City Attributes *This dataset contains 5 attributes covering NBA team cities' geographic information, demographic data, and economic indicators.*

Attribute	Description
city_name	Name of the city where NBA teams are located (e.g., Los Angeles, Boston)
state_name	Name of the state where the city is located (e.g., California, Massachusetts)
population	Total population of the city (e.g., 3971883)
area	Total area of the city in square kilometers; use decimal format (e.g., 1302.15)
gdp	Gross domestic product of the city in millions of dollars; use decimal format (e.g., 789543.50)

(7) Legal Dataset Attributes

This dataset contains 19 attributes documenting legal proceedings, including case participants, judicial decisions, financial penalties, and procedural details of court cases.

Attribute	Description
judge_name	name of the judge presiding over the case (e.g., Marshall J.)
plaintiff	name of the first person or entity who initiates a lawsuit or legal action (if there are multiple plaintiffs, extract only the first one mentioned; e.g., Cairnsmore Holdings Pty Ltd)
defendant	name of the first person or entity being sued or accused (if there are multiple defendants, extract only the first one mentioned; e.g., Narnia Investments Pty Ltd)
hearing_year	year when the hearing began; use format YYYY (e.g., 2009)
judgment_year	year when the judgment was delivered; use format YYYY (e.g., 2009)
charges	list of specific legal accusations or charges brought against the defendant (e.g., misleading conduct; price fixing; use a semicolon to separate multiple charges; leave empty if not applicable)
case_type	type of the case, choose one from [Criminal Case, Civil Case, Commercial Case, Administrative Case]; (e.g., Criminal Case)
verdict	court's decision, choose one from: [Guilty, Not Guilty, Others, Dismissed, Approved]; (e.g., Guilty)
legal_basis_num	Number of distinct legal statutes or codes cited in the judgment (e.g., 'Trade Practices Act 1974' counts as 1; enter an integer, e.g., 1)
case_number	Number of distinct precedent cases referenced in the judgment, including those cited, considered, discussed, or applied (e.g., 'Australian Competition and Consumer Commission v Rana' counts as 1; enter an integer, e.g., 1)
counsel_for_applicant	name of the applicant's lawyer (e.g., Mr I Faulkner SC)
counsel_for_respondent	name of the respondent's lawyer (e.g., Mr D Barclay)
nationality_for_applicant	applicant's nationality (e.g., Australia)
fine_amount	total monetary amount of fines imposed in the case (include all fines, partial payments, and fines for multiple parties; leave empty if none; e.g., 250000)
legal_fees	total amount of legal fees awarded or imposed (sum all applicable fees; leave empty if none; e.g., 30000)
plaintiff_current_status	current occupation or type of the plaintiff (if an individual, write occupation; if multiple people, use the first individual's occupation; if an organization, write 'organization'; if a company, write 'company'; if a government body, write 'government'; leave empty if unknown; e.g., company)
defendant_current_status	current occupation or type of the defendant (rules same as for plaintiff; e.g., government)
evidence	whether evidence was provided (1 if yes, 0 if none; e.g., 1)
first_judge	whether it was the first judgment (1 if yes, 0 if none; e.g., 1)

(8) Healthcare Dataset - Disease Attributes

This dataset contains 19 attributes providing comprehensive medical information about diseases, including their classification, clinical manifestations, diagnostic approaches, treatment options, and impact on patients' lives.

Attribute	Description
disease_name	The official name of the disease (e.g., Rheumatoid Arthritis)
disease_type	Comprehensive classification of the disease; choose one or more from ['autoimmune', 'infectious', 'genetic', 'metabolic', 'neoplastic', 'degenerative', 'congenital', 'traumatic', 'nutritional', 'vascular', 'inflammatory', 'endocrine', 'psychiatric', 'toxic', 'allergic', 'iatrogenic', 'idiopathic', 'parasitic', 'others'], separated by ' ' (e.g. neoplastic degenerative)
pathogenesis	The biological and physiological process by which the disease develops and progresses, choose one or more from: ['autoimmune', 'infectious_bacterial', 'infectious_viral', 'infectious_fungal', 'infectious_parasitic', 'genetic_mutation', 'metabolic_disorder', 'inflammatory', 'degenerative', 'neoplastic', 'vascular', 'traumatic', 'toxic', 'allergic', 'endocrine_disorder', 'nutritional_deficiency', 'iatrogenic', 'idiopathic', 'other'], separated by ' ' (e.g. inflammatory degenerative)
etiology	The underlying cause(s) of the disease, including genetic, environmental, infectious, or lifestyle factors (e.g., gene mutation, viral infection, smoking)
diagnostic_methods	Primary diagnostic approaches used for this disease; choose one or more from ['clinical_evaluation', 'laboratory_test', 'imaging', 'biopsy', 'genetic_testing', 'endoscopy', 'functional_test', 'pathology', 'screening', 'other'], separated by ' ' (e.g. clinical_evaluation laboratory_test)
common_symptoms	Typical clinical manifestations or symptoms experienced by patients with this disease (e.g., fever, cough, joint pain)
complications	Significant secondary conditions or health problems that can arise as a result of the disease (e.g., heart failure, kidney injury)
affected_organ	Primary organs, tissues, or body systems impacted by the disease (e.g., lungs, liver, joints, nervous system)
treatments	List of treatment modalities, choose one or more from ['oral_medication', 'injection', 'surgery', 'radiotherapy', 'chemotherapy', 'physical_therapy', 'psychotherapy', 'lifestyle_modification', 'device_implantation', 'dietary_intervention', 'others'], separated by ' ' (e.g. oral_medication)
drugs	List the specific names of drugs or pharmaceuticals used to treat or control the disease (e.g., metformin, aspirin, infliximab, penicillin)
prognosis	Typical course and expected outcome of the disease; choose one or more from ['full_recovery', 'partial_recovery', 'chronic_condition', 'remission_possible', 'high_recurrence_risk', 'low_recurrence_risk', 'high_mortality', 'moderate_mortality', 'low_mortality', 'disability', 'variable', 'stable', 'progressive', 'quality_of_life_impairment', 'asymptomatic', 'unknown'], separated by ' ' (e.g. partial_recovery chronic_condition)
sequelae	Long-term or permanent health problems, symptoms, or functional impairments that persist after the acute phase of a disease or injury has resolved (e.g., paralysis after stroke, chronic pain, hearing loss or cognitive impairment)

Attribute	Description
epidemiology	Population-level data about the disease, including prevalence, incidence, age/gender distribution, or notable epidemiological patterns (e.g., 1 in 1000 adults, more common in males)
risk_factors	Factors that increase the likelihood of developing the disease, choose one or more from ['family_history', 'genetic_predisposition', 'age', 'gender', 'smoking', 'alcohol_consumption', 'obesity', 'overweight', 'sedentary_lifestyle', 'unhealthy_diet', 'hypertension', 'hyperlipidemia', 'diabetes', 'chronic_disease', 'immunosuppression', 'environmental_exposure', 'occupational_exposure', 'exposure_to_toxins', 'infection', 'stress', 'low_socioeconomic_status', 'radiation_exposure', 'other'], separated by ' ' (e.g. family_history age)
preventive_measures	Known methods or actions to reduce the risk of disease occurrence, choose one or more from ['vaccination', 'lifestyle_modification', 'screening', 'prophylactic_medication', 'personal_hygiene', 'health_education', 'protective_equipment', 'environmental_control', 'vector_control', 'safe_sex_practices', 'regular_exercise', 'dietary_management', 'early_detection', 'other'], separated by ' ' (e.g. vaccination regular_exercise)
diagnosis_challenges	Typical difficulties or obstacles encountered in the diagnosis of the disease (e.g., nonspecific symptoms, lack of early biomarkers, disease mimics)
treatment_challenges	Major issues in the treatment or management of the disease (e.g., drug resistance, frequent relapse, side effects, poor adherence)
quality_of_life_impact	How the disease affects the patient's daily living, social interactions, work, and psychological well-being choose one or more from: ['mobility_impairment', 'chronic_pain', 'fatigue', 'sleep_disorders', 'anxiety', 'depression', 'social_isolation', 'work_disability', 'cognitive_impairment', 'sexual_dysfunction', 'self_care_difficulty', 'dietary_restriction', 'dependence_on_others', 'body_image_issues', 'financial_burden', 'emotional_distress', 'stigma', 'quality_of_life_decline', 'other'], separated by ' ' (e.g. chronic_pain work_disability)

(9) Healthcare Dataset - Drug Attributes

This dataset contains 17 attributes detailing pharmaceutical drugs, including their composition, administration methods, therapeutic uses, safety profiles, and storage requirements.

Attribute	Description
generic_name	The standard generic name(s) of the drug (e.g., Sildenafil with Dapoxetine)
brand_name	The commercial or brand name of the drug (e.g., Super Kamagra)
disease_name	The main disease(s) the drug is intended to treat; use standard disease names (e.g., Hypertension, Diabetes Mellitus, Asthma)
indication	The main symptoms, clinical manifestations, or related conditions for which the drug may be used (e.g., fever, pain, edema, high cholesterol, wheezing)
active_ingredients	Active pharmaceutical ingredients in the drug (e.g., Sildenafil Dapoxetine)
pharmaceutical_form	The physical form in which the drug is administered; choose one or more from [tablet, capsule, injection, solution, cream, ointment, powder, granule, suspension, patch, spray, suppository, other], separated by (e.g. tablet injection)

Attribute	Description
manufacturer	The name of the company or manufacturer producing the drug (e.g., Ajanta Pharma)
administration_route	The route through which the drug is administered; choose one or more from [oral, intravenous, intramuscular, subcutaneous, topical, transdermal, rectal, nasal, inhalation, other], separated by (e.g. oral other)
recommended_usage	Recommended instructions for use; choose one or more from [swallowed with water, chewable, dissolve in water, take with food, take without food, take with or without food, at fixed time, before meals, after meals, other], separated by (e.g. chewable other)
single_dose	The amount of the drug to be taken at one time per administration; extract all 'population + dose' pairs (e.g., adults 10 mg children 5 mg patients with EGPA 3x100 mg)
dosage_frequency	The recommended frequency of administration (e.g., once daily, twice a day, every 8 hours)
mechanism_of_action	How the drug works in the body to achieve its intended effect (e.g., improves blood circulation in penile vessels, supporting erection)
side_effects	Known or common side effects associated with the drug (e.g., photosensitivity, nausea, dizziness, drowsiness, tenderness)
activation_conditions	Specific conditions or requirements for the drug to take effect; choose one or more from [requires stimulation, requires food, requires fasting, requires physical activity, no special condition, requires specific timing, requires co-administration with other drugs, other], separated by (e.g. requires stimulation requires food)
prescription_status	Whether the drug requires a prescription or can be bought over-the-counter; choose one from [prescription_only, over_the_counter, restricted, unclassified]
unsuitable_population	Main populations for which the drug is not suitable or should be avoided; specify as described in the document (e.g., children, elderly, pregnant women, patients with renal impairment, children under 12)
storage_conditions	Recommended conditions for storing the drug; choose one or more from [store at room temperature, store in a cool place, refrigerate, freeze, protect from light, keep in a dry place, keep tightly closed, avoid freezing, other], separated by (e.g. avoid freezing other)

(10) Healthcare Dataset - Medical Institutes Attributes

This dataset contains 15 attributes describing medical research institutions, including their organizational structure, research focus areas, technological capabilities, and collaborative partnerships.

Attribute	Description
institution_name	Full name of medical research institution
institution_type	Type from: public, private, university-affiliated, corporate research lab
parent_organization	Parent or affiliated organization
establishment_year	Year institution was established
number_of_staff	Number of staff/researchers
leadership	Leading figures, directors, notable experts
institution_country	Country where located
institution_city	City where located
research_diseases	Main diseases focused on
research_fields	Areas from: molecular_biology, regenerative_medicine, immunology, genomics, pharmacology, oncology, neuroscience, bioinformatics, microbiology, epidemiology, clinical_research, biochemistry, cell_biology, virology, pathology, public_health, medical_imaging, others
key_technologies	Key technologies/methods/platforms used

Attribute	Description
key_achievements	Major research achievements/breakthroughs
international_collaboration	Partnerships with other institutions
funding_sources	Sources from: government grant, industry grant, philanthropy, self funded, international grant, other
technology_application	Practical applications of technologies
rating	the rating of this institution
expertise	the expertise of this institution
profit_status	the profit_status of this institution

(11) Finance Dataset Attributes

This dataset contains 25 attributes capturing comprehensive financial and corporate information, including company structure, financial performance metrics, shareholder information, and business risk assessments.

Attribute	Description
company_name	The full legal name of the company as disclosed in the annual report.
registered_office	The registered address or main office location of the company.
exchange_code	The code of the stock exchange where the company is listed (e.g., ASX, SUN).
principal_activities	Main business activities or industry classification; choose one or more from ['Mining', 'Finance', 'Healthcare', 'Manufacturing', 'Technology', 'Retail', 'Energy', 'Utilities', 'Real Estate', 'Transportation', 'Agriculture', 'Telecommunications', 'Media', 'Other'], separated by ' ' (e.g. Mining Finance).
board_members	Profiles of board members.
executive_profiles	Profiles of executive management team members.
revenue	Total operating revenue recognized during the reporting period. Enter a number. If the original currency is not USD, convert to USD according to the latest available exchange rate.
net_profit_or_loss	Net profit or loss amount for the reporting period (enter a negative number for loss). Enter a number. If the original currency is not USD, convert to USD according to the latest available exchange rate.
total_debt	Total debt outstanding at the end of the reporting period. Enter a number. If the original currency is not USD, convert to USD according to the latest available exchange rate.
total_assets	Total assets held by the company at the end of the reporting period. Enter a number. If the original currency is not USD, convert to USD according to the latest available exchange rate.
cash_reserves	Total cash and cash equivalents held at period-end. Enter a number. If the original currency is not USD, convert to USD according to the latest available exchange rate.
net_assets	Net asset value at the end of the reporting period. Enter a number. If the original currency is not USD, convert to USD according to the latest available exchange rate.
earnings_per_share	Earnings per share (EPS) for the reporting period. Enter a number. If the original currency is not USD, convert to USD according to the latest available exchange rate.
dividend_per_share	Dividends paid per share for the period. Enter a number. If the original currency is not USD, convert to USD according to the latest available exchange rate.
largest_shareholder	The name of the largest shareholder.
the_highest_ownership_stake	The highest percentage of ownership held by a single shareholder. Enter a number, unit: percent (%).
major_equity_changes	Whether there were major equity changes during the period; choose one from ['Yes', 'No'].
major_events	Type(s) of major events; choose one or more from ['M&A', 'Litigation', 'Major Contract', 'Leadership Change', 'Restructuring', 'Delisting', 'Other'], separated by ' ' (e.g. Litigation).

Attribute	Description
business_sales	Total operating revenue from all major business lines or segments. If the document reports revenue for multiple segments separately, sum all segments to obtain the total. Enter a single number. If the original currency is not USD, convert to USD according to the latest available exchange rate.
business_profit	Total profit contributed by all major business lines or projects. If the document reports profit for multiple segments separately, sum all segments to obtain the total. Enter a single number. If the original currency is not USD, convert to USD according to the latest available exchange rate.
business_cost	Total cost associated with all main business lines or projects. If the document reports cost for multiple segments separately, sum all segments to obtain the total. Enter a single number. If the original currency is not USD, convert to USD according to the latest available exchange rate.
business_segments_num	Number of business segments reported. Enter a number.
business_risks	Main types of business risks; choose one or more from ['Market Risk', 'Credit Risk', 'Operational Risk', 'Legal/Compliance Risk', 'Environmental Risk', 'Strategic Risk', 'Other'], separated by ' ' (e.g. Credit Risk Market Risk).
remuneration_policy	Executive remuneration policy type; choose only one from ['Fixed', 'Performance-based', 'Stock Option', 'Mixed', 'Not Disclosed'].
auditor	Name of the external audit firm.

A.3 Evaluation Results

Table 6 and 7 present the average precision, recall, and F1-score of all queries across all the datasets. Given a query Q , the set of tuples returned by a method is denoted as $T(Q)$, we compute the precision, recall, and F1-score for each column individually. The average of these column-level scores is then reported as the overall precision, recall, and F1-score for the corresponding SQL query. Table 6 shows the experimental results of different methods on queries with only extraction. And Table 7 shows the experimental results of different methods on queries with filter.

A.4 Ablation Study of Chunking Strategies

In our ablation studies, several text chunking strategies are utilized to divide long documents for subsequent processing tasks. In this study, we consider four representative chunking methods: fixed-length chunking, heuristic chunking, semantic chunking, and tree-based chunking. The implementation details of each strategy are provided in the following paragraphs.

Fixed-length chunking splits the text into segments of a fixed length (e.g., 512 tokens), with an overlap of 128 tokens between adjacent chunks to reduce information loss in long-text scenarios. This method is simple and fast to process, but it often breaks complete sentences, resulting in information fragmentation and semantic incompleteness.

Heuristic chunking divides the text recursively based on the priority of delimiters (such as '\n', '.', '!', '?'), ensuring that each chunk contains no more than 512 tokens. Higher-level delimiters are preferred, and when the chunk size constraint cannot be satisfied, the method falls back to the next delimiter level. This strategy minimizes sentence splitting and preserves information integrity, but its effectiveness depends on the original text format and paragraph segmentation.

Semantic chunking (e.g., QUEST) first applies sentence splitters (such as spaCy or nltk) to segment the text into sentences, and then merges semantically similar adjacent sentences using the semantic chunking module in LangChain. A sliding window calculates the average semantic similarity, and a new chunk is created whenever the similarity exceeds a predefined threshold (typically set at the 95th percentile of all pairwise sentence similarities). Chunk lengths are restricted to between 128 and 512 tokens to balance recall and embedding quality. This approach better preserves semantic continuity and information richness, thereby improving embedding quality, but it incurs higher computational cost and slower processing speed.

Tree-based chunking (e.g., ZenDB) utilizes the hierarchical structure of document headings by constructing an SHT tree. The content between every two headings is assigned to a node, and a summary for each section is generated via post-order traversal to facilitate retrieval. This method is particularly suitable for PDF and Markdown documents with clear hierarchical structures: Markdown headings are extracted using regular expressions, and PDF headings are identified via pattern matching (see [20] for details). Tree-based chunking effectively avoids sentence fragmentation and preserves more tokens and context, but introduces increased chunking and processing overhead, resulting in greater latency.

Method	Art			Player			Legal		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
Fixed-length	0.50	0.56	0.49	0.60	0.80	0.67	0.41	0.44	0.40
Heuristic	0.58	0.57	0.55	0.64	0.79	0.69	0.50	0.44	0.44
Semantic	0.59	0.58	0.56	0.68	0.80	0.72	0.52	0.48	0.47
Tree	0.51	0.56	0.50	0.66	0.79	0.70	0.49	0.45	0.45

Table 10: Ablation Study of Chunking Strategies.

A.5 Evaluation Prompts

To conduct our evaluation, we designed prompts for each baseline according to the implementation design of different operators in each baseline. Here we present part of the necessary prompt for each.

PZ Prompts:

(1) convert

```
COT_QA_BASE_SYSTEM_PROMPT = """You are a helpful
assistant whose job is to {job_instruction}.
You will be presented with a context and a set of
output fields to generate. Your task is to
generate a JSON object that fills in the output
fields with the correct values.
You will be provided with a description of each
input field and each output field. All of the
fields in the output JSON object can be derived
using information from the context.
{output_format_instruction} Finish your response
with a newline character followed by ---
An example is shown below:
---
INPUT FIELDS:
{example_input_fields}
OUTPUT FIELDS:
{example_output_fields}
CONTEXT:
{example_context}
{image_disclaimer}
Let's think step-by-step in order to answer the
question.
REASONING: {example_reasoning}
ANSWER:
{example_answer}
---
"""
```

```
COT_QA_BASE_USER_PROMPT = """You are a helpful
assistant whose job is to {job_instruction}.
You will be presented with a context and a set of
output fields to generate. Your task is to
generate a JSON object which fills in the
output fields with the correct values.
You will be provided with a description of each
input field and each output field. All of the
fields in the output JSON object can be derived
using information from the context.
{output_format_instruction} Finish your response
with a newline character followed by ---
"""
```

Method	Art			CSPaper			Player			Legal			Finance			Healthcare		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
Evaporate	0.33	0.33	0.33	0.38	0.38	0.38	0.34	0.34	0.34	0.31	0.32	0.31	0.21	0.33	0.24	0.10	0.11	0.10
Palimpzest	0.61	0.65	0.63	0.65	0.65	0.65	0.88	0.88	0.88	0.68	0.67	0.67	0.53	0.51	0.52	0.24	0.24	0.24
LOTUS	0.58	0.63	0.60	0.64	0.64	0.64	0.86	0.87	0.87	0.66	0.66	0.66	0.51	0.54	0.52	0.24	0.25	0.24
DocETL	0.51	0.51	0.51	0.56	0.56	0.56	0.89	0.89	0.89	0.57	0.57	0.57	0.43	0.44	0.43	0.17	0.18	0.18
ZenDB	0.47	0.47	0.47	0.58	0.58	0.58	0.82	0.84	0.83	0.57	0.57	0.57	0.49	0.48	0.48	0.24	0.24	0.24
QUEST	0.45	0.55	0.48	0.60	0.60	0.60	0.84	0.84	0.84	0.58	0.58	0.58	0.50	0.49	0.49	0.56	0.23	0.32
UQE	0.57	0.57	0.57	0.62	0.62	0.62	0.86	0.87	0.86	0.65	0.66	0.65	0.51	0.53	0.52	0.23	0.22	0.22

Table 6: Performance Comparison of Queries with only Extraction.

Method	Art			CSPaper			Player			Legal			Finance			Healthcare		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
Evaporate	0.07	0.01	0.01	0.27	0.27	0.27	0.31	0.12	0.16	0.13	0.11	0.12	0.06	0.18	0.08	0.03	0.01	0.01
Palimpzest	0.32	0.33	0.29	0.57	0.65	0.56	0.66	0.77	0.69	0.54	0.55	0.54	0.36	0.46	0.39	0.22	0.41	0.26
LOTUS	0.24	0.53	0.29	0.51	0.52	0.52	0.48	0.71	0.56	0.49	0.62	0.53	0.23	0.51	0.37	0.19	0.34	0.23
DocETL	0.23	0.23	0.23	0.45	0.45	0.45	0.70	0.79	0.73	0.55	0.57	0.55	0.31	0.32	0.29	0.21	0.40	0.23
ZenDB	0.21	0.23	0.22	0.38	0.67	0.43	0.72	0.64	0.68	0.49	0.44	0.43	0.34	0.40	0.35	0.18	0.47	0.23
QUEST	0.21	0.25	0.22	0.42	0.70	0.47	0.64	0.79	0.70	0.50	0.42	0.44	0.35	0.42	0.36	0.20	0.58	0.27
UQE	0.19	0.19	0.19	0.38	0.38	0.38	0.55	0.73	0.62	0.37	0.36	0.36	0.22	0.59	0.30	0.18	0.23	0.16

Table 7: Performance Comparison of Queries with Filter.

Method	Player			Healthcare		
	Precision	Recall	F1	Precision	Recall	F1
Extract-All	0.57	0.77	0.64	0.15	0.33	0.18
with-Search	0.50	0.83	0.62	0.12	0.41	0.17

Table 8: Evaluation of Join Optimization.

Query	Method	Art			Player			Legal		
		Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
Extract	E5	0.46	0.46	0.46	0.83	0.83	0.83	0.53	0.53	0.53
	Text	0.53	0.53	0.53	0.84	0.84	0.84	0.55	0.55	0.55
	Qwen	0.55	0.53	0.55	0.89	0.89	0.89	0.58	0.58	0.58
Filter	E5	0.40	0.41	0.40	0.61	0.76	0.67	0.41	0.43	0.40
	Text	0.40	0.44	0.41	0.64	0.77	0.68	0.37	0.44	0.40
	Qwen	0.42	0.50	0.44	0.64	0.79	0.70	0.48	0.46	0.44

Table 9: Ablation Study of Embedding Models.

Method	Extract			Filter		
	Precision	Recall	F1	Precision	Recall	F1
gpt-4.1-mini	0.58	0.58	0.58	0.50	0.44	0.44
gpt-4.1-nano	0.52	0.52	0.52	0.42	0.42	0.40
gpt-4.1	0.63	0.62	0.63	0.53	0.48	0.48
DeepSeek-V3	0.61	0.61	0.61	0.53	0.45	0.45
claude-sonnet-4-20250514	0.61	0.61	0.61	0.51	0.49	0.46

Table 11: Ablation Study of LLMs.

```

INPUT FIELDS:
{input_fields_desc}
OUTPUT FIELDS:
{output_fields_desc}
CONTEXT:
{context}
<<image-placeholder>>
Let's think step-by-step in order to answer the
question.
REASONING: ""
(2) filter
COT_BOOL_BASE_SYSTEM_PROMPT = ""You are a helpful
assistant whose job is to {job_instruction}.
You will be presented with a context and a filter
condition. Output TRUE if the context satisfies
the filter condition, and FALSE otherwise.
Remember, your answer must be TRUE or FALSE. Finish
your response with a newline character followed
by ---
An example is shown below:
---
INPUT FIELDS:
{example_input_fields}
CONTEXT:
{example_context}
{image_disclaimer}
FILTER CONDITION: {example_filter_condition}
Let's think step-by-step in order to answer the
question.
REASONING: {example_reasoning}
ANSWER: TRUE

```

```

---
"""

COT_BOOL_BASE_USER_PROMPT = """You are a helpful
assistant whose job is to {job_instruction}.
You will be presented with a context and a filter
condition. Output TRUE if the context satisfies
the filter condition, and FALSE otherwise.
Remember, your answer must be TRUE or FALSE. Finish
your response with a newline character followed
by ---

---
INPUT FIELDS:
{input_fields_desc}
CONTEXT:
{context}
<<image-placeholder>>
FILTER CONDITION: {filter_condition}
Let's think step-by-step in order to answer the
question.
REASONING: """

```

LOTUS Prompts:

(1) extract

```

prompt = "What" + {attribute} + "in {context}?" + {
    attribute's description} + "If there are
multiple values, separate them with '|' and
leave empty if not applicable. Please keep each
extracted value concise and avoid lengthy
content."

```

(2) filter

```

prompt = "{context}." + {attribute1's description} +
{attribute2's description} + {filters}

```

DocETL Prompts:

A pipeline in DocETL consists of five main components:

- **Default Model:** The language model to use for the pipeline.
- **System Prompts:** A description of your dataset and the "persona" you'd like the LLM to adopt when analyzing your data.
 - Dataset Description: A concise explanation of what kind of data the model will be processing.
 - Persona: The role or perspective the model should adopt when analyzing the data.
- **Datasets:** The input data sources for your pipeline.
- **Operators:** The processing steps that transform your data.
- **Pipeline Specification:** The sequence of steps and the output configuration.
 - Steps: The sequence of operations to be applied to the data.
 - Output: The configuration for the final output of the pipeline.

Example Operators:

(1) Map Operator

```

- name: analyze_news_article
  type: map
  prompt: |
    Analyze the following news article:

```

```

"{{ input.article }}"

```

Provide the following information:

1. Main topic (1-3 words)
2. Summary (2-3 sentences)
3. Key entities mentioned (list up to 5, with brief descriptions)
4. Sentiment towards the main topic (positive, negative, or neutral)
5. Potential biases or slants in reporting (if any)
6. Relevant categories (e.g., politics, technology, environment; list up to 3)
7. Credibility score (1-10, where 10 is highly credible)

output:

```

schema:
  main_topic: string
  summary: string
  key_entities: list[object]
  sentiment: string
  biases: list[string]
  categories: list[string]
  credibility_score: integer

```

model: gpt-4o-mini

(2) Filter Operator

```

- name: filter_high_impact_articles
  type: filter
  prompt: |
    Analyze the following news article:
    Title: "{{ input.title }}"
    Content: "{{ input.content }}"

```

Determine if this article is high-impact based on the following criteria:

1. Covers a significant global or national event
2. Has potential long-term consequences
3. Affects a large number of people
4. Is from a reputable source

Respond with 'true' if the article meets at least 3 of these criteria, otherwise respond with 'false'.

output:

```

schema:
  is_high_impact: boolean

```

model: gpt-4-turbo

```

validate:
  - isinstance(output["is_high_impact"], bool)

```

(3) Reduce Operator

```

- name: summarize_feedback
  type: reduce
  reduce_key: department
  prompt: |
    Summarize the customer feedback for the {{
      inputs[0].department }} department:

    {% for item in inputs %}
    Feedback {{ loop.index }}: {{ item.feedback }}
    {% endfor %}

    Provide a concise summary of the main points and
    overall sentiment.
  output:
    schema:
      summary: string
      sentiment: string
(4) Equijoin Operator
- name: match_candidates_to_jobs
  type: equijoin
  comparison_prompt: |
    Compare the following job candidate and job
    posting:

    Candidate Skills: {{ left.skills }}
    Candidate Experience: {{ left.years_experience
    }}

    Job Required Skills: {{ right.required_skills }}
    Job Desired Experience: {{ right.
    desired_experience }}

    Is this candidate a good match for the job?
    Consider both the overlap
    in skills and the candidate's experience level.
    Respond with "True" if
    it's a good match, or "False" if it's not a
    suitable match.
  output:
    schema:
      match_score: float
      match_rationale: string

```

ZENDB Prompts:

(1) Syntax Tree Constructing

```

prompt = (
    "You are analyzing a document. "
    "Section headers in a document typically
    indicate the start of a new section or
    subsection, such as chapter titles, main
    headings, or important topic divisions. "

```

"Section headers are often short, descriptive, and may be formatted differently from the main text. "

"Given the following phrase from the document:\n\n"

f'"{phrase}"\n\n'

"Is this phrase a section or subsection header in the document? "

"Answer only 'yes' or 'no'."

(2) Node Retrieval

prompt = f'""You are given a list of sections from a document like a Art artist, biographical article and so on, each with a title and a summary.

Your task is to select the section most likely to contain data for the table "{table_name}".

Table description: "{description}"

Table fields: {fields_str}

Sections:

{chr(10).join(node_info)}

Return ONLY the node ID (e.g., "doc_0_node_5") that best matches the table content, or "NONE" if no match.

""

prompt = f'""Check if this academic paper section might contain data about: {conditions_text}

Section Title: {node.name}

Section Summary: {node.summary[:200]}

Section Content: {node.full_context[:300]}

Answer ONLY 'yes' or 'no'""

(3) Filter

if operator in ['=', '==']:

prompt = f'""Check if the {field} in this text equals "{value}":\n\n{node.full_context[:500]}\n\nAnswer ONLY 'yes' or 'no'""

elif operator == 'LIKE':

prompt = f'""Check if the {field} in this text contains or is similar to "{value}":\n\n{node.full_context[:500]}\n\nAnswer ONLY 'yes' or 'no'""

elif operator == '>':

prompt = f'""Check if the {field} in this text is greater than {value}:\n\n{node.full_context[:500]}\n\nAnswer ONLY 'yes' or 'no'""

elif operator == '<':

prompt = f'""Check if the {field} in this text is less than {value}:\n\n{node.full_context[:500]}\n\nAnswer ONLY 'yes' or 'no'""


```

elif operator == '>=':
    prompt = f"""Check if the {field} in this text
    is greater than or equal to {value}:\n\n{node.
    full_context[:500]}\n\nAnswer ONLY 'yes' or 'no'
    """
elif operator == '<=':
    prompt = f"""Check if the {field} in this text
    is less than or equal to {value}:\n\n{node.
    full_context[:500]}\n\nAnswer ONLY 'yes' or 'no'
    """
elif operator == '!=':
    prompt = f"""Check if the {field} in this text
    is NOT equal to "{value}":\n\n{node.
    full_context[:500]}\n\nAnswer ONLY 'yes' or 'no'
    """
else:
    return True

```

(4) Attribute Value Extraction

```

prompt = f"""
You are analyzing a document. Please extract the
content of the "{field}" field from the text.
- If the field is not found in the text, return
"NULL".
The original text is as follows:
{node.full_context[:1000]}
Only return the content of the field itself, and
do not include any additional information.
"""

```

QUEST Prompts:

(1) Sampling

User Prompt:

Your Task is to extract key-value pairs from text chunks with following guides:

1. Input:
 - Schema: Attributes to be extracted and their corresponding descriptions
 - Chunks: A list of text chunks to be extracted, each marked with its ID at the beginning.
2. Output:
 - `key`: lowercase attribute_name from schema (e.g., name)
 - `value`: attribute_value with exact casing / spacing (e.g., iPhone 14)
 - `confidence`: int, between 0 to 100.
 - `chunkid`: int, id of the chunk from which the key-value pair is extracted.
 - Output one tuple per line, formatted as (attr_name, attr_value, confidence, chunkid).

Schema: ```{attr_schema}```

Document: ```{doc}```

System Prompt:

You are an attribute extraction assistant. Only respond with (key, value, confidence, chunkid) pairs. Do not include any explanations or extra text.

(2) Extract

User Prompt:

Extract the following fields from the given document : {attributes}.

Instructions:

- Format your response as lines, each in the format: `field: value`
- Use the exact field names: {attributes}
- Follow the descriptions of the fields: ``` {related_attr_descriptions_str} ```
- If a field is missing or unknown, leave its value empty (e.g., `team: None`)
- use the line break (`\n`) to split the lines
- Do not add any extra text, comments, or explanations

Document: ```{doc}```

System Prompt:

You are an information extraction assistant. Only respond with key-value pairs using the exact field names provided. Do not include any explanations or extra text.

(3) Filter

User Prompt:

Check if the value satisfy the condition or semantically similar to the condition.

Instructions:

- Format your response as a single line, in the format: `fcondition: True/False`.
- Check if the value satisfies the condition {filters}.
- The filter condition `==` or `IN` can be considered emantically for strings. For example , `Lakers\` and `Los Angeles Lakers\` are equal, `fashion\` and `Fashion || Illustration\` are also equal.
- The filter condition `<`, `>`, `>=`, `<=`, can be considered as numeric comparison or a date comparison, note that the eariler date is smaller.
- Do not add any extra text, comments, quotes, or explanations.

Value: ```{value}```

Condition: ```{filters}```

System Prompt:

You are an condition check assistant. Respond in a single line, includes a pair format as `fcondition: True/False`, the boolean True or False represents whether the condition is met. Do not include any explanations or extra text.

(4) Extract + Filter

User Prompt:

Extract the following field from the given document: {attributes}. Then Check if the value satisfy the condition.

Instructions:

- Format your response as two lines, the first line in the format: `field: value`, and the second line in the format: `fcondition: True/False`.
- Use the exact field name: {attributes}.
- Check the condition {filters}.
- If the field is missing or unknown, leave its value empty (e.g., `team: None`), and leave the condition as False.
- use the line break (`\n`) to split the lines.
- You should first extract the field value and then check. For example, we first do extract, and get filed is `name`, value is `Lee`. Then we do check, the condition is `name=='Frank'`, the value does not satisfy the condition, so the fcondition is False.
- The filter condition `==` or `IN` can be considered emantically for strings. For example, `Lakers` and `Los Angeles Lakers` are equal, `fashion` and `Fashion || Illustration` are also equal.
- The filter condition `<`, `>`, `>=`, `<=` can be considered as numeric comparison or a date comparison, note that the eariler date is smaller.
- Follow the descriptions of the field:
``` {related\_attr\_descriptions\_str} ```
- Do not add any extra text, comments, quotes, or explanations.

Document: ```{doc}```

*System Prompt:*

You are an information extraction and check assistant. Respond in two lines, the first line is a key-value pair using the exact field name provided; the second line is a key-value pair, and value is a boolean True or False whether the condition is met. Do not include any explanations or extra text.

#### UQE Prompts:

##### (1) Extract Prompt

```
system_prompt["description"]["extract"] =
"You are an art analyst. The following paragraph " +
"provides some basic information about the artist. *
info_intro* " +
"Please Help me to extract the value about *
attrs_in_prompt*. " +
"You should present the result in the form of 'attr:
value' and " +
"connect different pairs with separator '&&'. " +
"For example, 'attr1: 5&&attr2: public&&...'. If the
value of the " +
"attr is not available, please return 'None'."
```

##### (2) Filter Prompt

```
system_prompt["description"]["filter"] =
"You are an art analyst. The following paragraph " +
"provides some basic information about the company.
info_intro " +
"Please Help me to extract the information about *
attr_name* " +
"and determine whether the it satisfies '*attr_name*
cond_op *cond_val*'. " +
"If it does, please only return True. Otherwise,
please return False. " +
"Please only return 'True' or 'False'."
```

##### (3) GroupBy Labels Split Prompt

```
system_prompt["description"]["groupby_labels_split"]
=
"I will provide an introduction of the artist. *
info_intro* " +
"Please help me to extract the attribute '*attr_name
*'. " +
"You should only present the result, without any
other information."
```

##### (4) GroupBy Class Prompt

```
system_prompt["groupby_class"] =
"I will provide you several paragraphs with their
column names in the " +
"following conversation. " +
"Also, I will provide a list of attributes,
including the possible " +
"categories for each attribute " +
```

"in the form of '<column>.<attribute>:<list of categories>'. " +  
 "The attributes and their candidate categories are: \*attr\_with\_labels\*. " +  
 "Please extract the attribute from the corresponding paragraph and " +  
 "classify them into the given categories. " +  
 "You should present the classification result of each attributes in the " +  
 "form of 'attr1:category1&&attr2:category2&&...' " +  
 "If none of the given category is matched for the given attribute, " +  
 "please return 'attr:None' as its classification result." +  
 "You should only return the results, without any other information."

#### (5) Merge Filter Prompt

```
system_prompt["merge_filter"] =
```

"I will provide you several paragraphs in the following conversation. " +  
 "Each paragraph is preceded by a name indicator. " +  
 "\*col\_list\* " +  
 "Please help me to filter the paragraphs according to the given conditions. " +  
 "The conditions on a paragraph are presented in the form of " +  
 "'<name1>:<condition1>,<condition2>,<condition3>,<...>'. " +  
 "Each condition is presented in the form of < attr\_name> <cond\_op> <cond\_val>. " +  
 "You should first extract the information of < attr\_name> from the paragraph " +  
 "and then determine whether the condition is satisfied. " +  
 "The conditions are: \*condition\_list\*. " +  
 "The explanation on the attributes included in the conditions are: " +  
 "\*info\_intro\*" +  
 "You only need to give an overall decision on all these paragraphs and " +  
 "conditions. " +  
 "Return 'True' if all the conditions are satisfied " +  
 "+  
 "and return 'False' if some of the conditions are not satisfied. " +  
 "Please only return the result 'True' or 'False', without other information."

#### (6) Merge Filter Aggregation Split Prompt

```
system_prompt["merge_filter_aggr_split"] =
```

"I will provide you several paragraphs in the following conversation. " +  
 "Each paragraph is preceded by a name indicator. " +  
 "Please help me to filter the paragraphs according to the given condition. " +

"The conditions to be satisfied are: \* filter\_condition\* " +  
 "Each condition is presented in the form of (<name1>.<attribute1> <op> <value1>). " +  
 "The explanation on the attributes included in the conditions are: " +  
 "\*info\_intro\*" +  
 "Return 'True' if all the conditions are satisfied " +  
 "+  
 "and return 'False' otherwise. Please Only return ' True' or 'False', " +  
 "without any other information."

#### Additional Indicators:

```
system_prompt["description"]["
```

```
merge_filter_column_indicators"] = "The
```

```
paragraph 'description' is a description of the
```

```
artist in text form."
```

```
system_prompt["description"]["
```

```
merge_filter_cond_indicators"] =
```

```
"The conditions on paragraph 'description' include:
```

```
[*col_conditions*]"
```